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Requirement Analysis and Specification Document Students&Companies

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1 | Introduction

1.1. Purpose

Finding the right company in which to carry out internships is an issue that a lot of students must handle, both to complete their university carrier and to firstly approach the workplace environment. Unfortunately, it is not always easy for students to find a company whose projects and activities match their interests and skills, especially when the number of available opportunities is high. From another point of view, companies must deal every day with the selection of CVs, basing on students' attitudes and experiences, in order to find suitable figures for the activities they offer.

Students&Companies (S&C) is a platform that helps the matching between university students looking for internships and companies that offer these opportunities. The main idea is that students can publish their CV on the platform to find internships that might interest them, while companies advertise their internships so that the system can help them find suitable students. Companies can also set up and plan interviews with the matching students through the S&C platform, which helps them in the selection process by collecting and analyzing students' answers through structured questionnaires. S&C enables also the interested parts to monitor, keep track and provide information about the execution of the matchmaking processes and the status of the related internships. The main goals of the platform are listed in the following section.

1.1.1. Goals

[G1] Students want to look for available internships that might interest them.

[G2] Companies would like to look for suitable students' CVs for the internships they offer.

[G3] Companies would like to admit suitable students to their internships through interviews and structured questionnaires.

[G4] Both students and companies would like to provide information to the other party

regarding the matchmaking process and the related internships (such as feedback, suggestions or complaints).

1.2. Scope

In this section, we will identify and describe S&C platform **domain**, by specifying world and shared phenomena. The two main involved actors are students and companies.

Students can upload their CV on the platform and can both search for available internships and explore the ones recommended by the system. They can initiate the matchmaking process by selecting an internship of their interest and they can also conclude it by choosing one for which a company has selected their CV. After a match is completed, students undergo an interview with the company and fill out a structured questionnaire to help the company reason about their selection. If they are admitted, students carry out an internship with the matching company and can communicate with it, providing feedback, suggestions and complaints regarding the matchmaking process or the status of the ongoing program.

Companies can advertise their internship offers on the platform and can view uploaded student CVs that are recommended by the system. Symmetrically, they can initiate the matchmaking process for a specific internship by selecting a CV that corresponds to their needs, and they can also conclude it by choosing a student that has previously showed interest in that internship. Companies then use the platform to set up interviews with matching students and to create structured questionnaires for them, so that the system can grade and rank students to facilitate the selection process. Lastly, based on what they have achieved in the previous step, companies can decide to admit or reject matching students for the internship programs. Similarly to students, companies are also able to communicate with those of them who are enrolled for their internships.

1.2.1. World Phenomena

[WP1] A student creates a CV containing their experiences, skills and attitudes.

[WP2] A company defines an internship by creating a document containing related projects and terms.

[WP3] A company interviews a student for an internship using an external virtual meeting platform.

[WP4] A student carries out an internship with the company that admitted them.

[WP5] A company schedules a virtual meeting for an interview using an external platform, obtaining a web link for it.

1.2.2. Shared Phenomena

World Controlled

[SP1] A student uploads their CV on the platform.

[SP2] A student removes their CV from the platform.

[SP3] A student selects an internship of their interest.

[SP4] A student is notified when a company selects their CV.

[SP5] A student is notified when an interview with them has been programmed by a company for an internship.

[SP6] A student is notified when a questionnaire for the selection process of an internship has been uploaded by the corresponding company.

[SP7] A student fills out a questionnaire for the selection process of an internship.

[SP8] A student is notified when a company admits them (or rejects them) after the selection process for an internship.

[SP9] A student provides feedback about an internship and its matching process to the corresponding company.

[SP10] A student fills out a satisfaction questionnaire for an internship or its matching process.

[SP11] A company publishes an internship on the platform by uploading the related document.

[SP12] A company remove an internship (i.e. and the related document) from the platform.

[SP13] A company selects a student CV that corresponds to its needs.

[SP14] A company is notified when a student selects one of its internships.

[SP15] A company sets up an interview with a student for one of its internships.

[SP16] A company creates a questionnaire for the selection process of the students for one of its internships using the platform.

[SP17] A company is notified when a student fills out the questionnaire for the selection process of one of its internships.

[SP18] A company admits a student for an internship after its selection process.

[SP19] A company rejects a student for an internship after its selection process.

[SP20] A company provides feedback about an internship and its matching process to the corresponding student.

[SP21] A company fills out a satisfaction questionnaire for an internship or its matching process with a student.

[SP22] After a company set up an interview with a student, the system adds the interview to both student's and company's calendars.

Machine Controlled

[SP23] The system recommends an internship to a student.

[SP24] The system notifies a student of a new available satisfaction questionnaire about an internship or its matching process.

[SP25] The system recommends a student CV to a company for one of its internships.

[SP26] The system notifies a company of a new available satisfaction questionnaire about an internship or its matching process with a student.

[SP27] The system evaluates a questionnaire of a student for their selection process for an internship, making the grade available to the corresponding company.

1.3. Definitions, Acronyms, Abbreviations

1.3.1. Definitions

- **Users:** they include both students and companies.
- **Matchmaking process:** phase in which students and companies show interest for each other in the context of an internship, possibly basing on system recommendations.
- **Selection process:** phase in which companies decide the students to admit for their internships, basing on interviews and questionnaires outcomes.
- **Credentials:** user e-mail address and password.

1.3.2. Acronyms

- **S&C:** Students&Companies.
- **API:** Application Programming Interface.
- **UI:** User Interface.
- **CV:** Curriculum Vitae.

1.3.3. Abbreviations

- **G*:** goal number *.
- **WP*:** world phenomenon number *.
- **SP*:** shared phenomenon number *.
- **R*:** functional requirement number *.
- **UC*:** use case number *.

1.4. Revision Hystory

Versions of the document.

1.5. Reference Documents

This document is mainly based on the following materials:

- RASD and DD assignment specification for the Software Engineering 2 course, a.y. 2024-2025.
- Slides of the Software Engineering 2 course, available on WeBeep.

1.6. Document Structure

To provide a better understanding of the document, we report its chapter division below:

1. **Introduction:** it briefly presents the project, describing the goals to be achieved and the main phenomena of the domain. It also provides some useful details for the understanding of the document.
2. **Overall Description:** it includes some possible scenarios and further information

about the domain. It also illustrates the main functionalities of the platform, as well as made assumptions.

3. **Specific Requirements:** it contains a comprehensive analysis of both functional and non functional requirements, in addition to constraints and relevant attributes of the system.
4. **Formal Analysis Using Alloy:** it describes the formal modeling activity performed using the Alloy tool.
5. **Effort Spent:** it shows the time spent by each team member for the fulfillment of the project.
6. **References:** here additional references to the main mentioned above are listed.

2 | Overall Description

2.1. Product Perspective

2.1.1. Scenarios

Student uploads a CV

Bob is a Computer Science and Engineering student, and he wants to carry out an internship in order to conclude his university carrier. Bob opens the S&C website and registers as “Student”, providing his personal information (complete name, gender, date and place of birth, address and name of the university in which he’s enrolled), his personal e-mail and a password; he is then redirected to an external platform, where he can authenticate with his institutional credentials; after registration, he logs-in to the platform with the same e-mail and password. Once in the S&C homepage, Bob goes to the section “Upload new CV”, attaches his personal CV document and confirms its upload on the platform.

Company publishes an internship document

TechForma Engineering S.R.L. (TF) is a company that wants to advertise a new internship offer in the field of Software Engineering (Int1), in order to find suitable university students to carry it out. TF’s Recruiter Alice opens the S&C website and registers as “Company”, providing some identification information (name of the company, type of business, name of the authorized representative, business registration number and address of the headquarter), the e-mail address of the Recruiting Department and a password; after registration (and verification of the company identity), she logs-in to the platform with the same e-mail and password. Once in the S&C homepage, Alice goes to the section “Publish new internship”, inserts a proper title and attaches a document containing a detailed description of the projects involved in the internship and its related terms, such as possible future job opportunities and wage. Lastly, Alice confirms the publication of the document on the platform; as a consequence, the system creates a section “Int1” in TF’s homepage.

Student selects a recommended internship

Student Bob, who has a profile on the S&C website and has already uploaded his CV, wants to find an internship offer that could interest him and that possibly aligns with his university carrier. He logs-in to the platform with his credentials and goes to the section “Recommended internships”, where he can view a list of available offers recommended by the system with their related documents. Bob reads the description document for Int1 and, since the related projects interest him, he selects it. As a consequence, the system shows the chosen internship in the section “Selected internships” of Bob’s S&C homepage and notifies the company TF about Bob’s selection, additionally showing Bob’s CV in the subsection “Interested students” of the section “Int1” in TF’s homepage.

Company selects a recommended CV

TF’s Recruiter Alice, which has registered the company on the website and has already published some internships on the platform (including Int1), wants to find suitable students to admit to Int1. She logs-in to the platform with TF’s credentials and, once entered the section dedicated to that internship, she goes to the section “Recommended students”, where she can view a list of students’ CVs recommended by the system. Alice reads Bob’s CV and decides to consider him for an interview, so she selects the CV. As a consequence, the system shows the chosen CV in the section “Selected students” for Int1 and notifies Bob about Alice’s selection, additionally showing the description document for Int1 in the section “Interested companies” of Bob’s homepage.

Student completes the matchmaking process

Student Bob wants to check if any company has shown interest in his CV for an internship offer. He enters the section “Interested companies” of his S&C homepage, where it’s shown that company TF has selected his CV for Int1. Bob reads the description document of the internship and decides to select it, completing the matchmaking process. As a consequence, the system notifies TF about the created match and shows Bob’s CV in the section “Completed matches” for Int1. In addition, also Bob can now view Int1 in the “Completed matches” section of his S&C homepage.

Company completes matchmaking process

TF’s Recruiter Alice wants to check if any student has shown interest in the company’s offer for Int1. She enters the section dedicated to that internship on TF’s homepage and goes into the subsection “Interested students”, in which is shown that Bob has selected their internship. Alice reads Bob’s CV and decides to select it, completing the match-

making process. As a consequence, the system notifies Bob about the created match and shows all the information regarding Int1 in the section “Completed matches” of his homepage. In addition, the system adds Bob’s CV and related information in the corresponding “Completed matches” section for Int1, in TF’s homepage.

Users fill out the satisfaction questionnaires regarding the matchmaking process

TF’s Recruiter Alice enters the section "Satisfaction questionnaires", where for each completed match and for each active internship a satisfaction questionnaire created by the system is available (only if this has not been filled out yet). Alice provides her opinion about the matchmaking process with student Bob by filling out the related questionnaire and submits it. In a second moment, Bob repeats the same process by filling out and submitting the satisfaction questionnaire related to the matchmaking process with Int1. After both questionnaires have been submitted, the system evaluates positively or negatively the related match based on users’ responses, in order to feed statistical analysis for future recommendation processes.

Company sets up an interview with a student

The scheduled period for Int1 is approaching and company TF needs to decide which students to admit. Alice enters the section dedicated to Int1, goes to the subsection “Completed matches” and starts the selection process by considering student Bob; therefore, she clicks the button “Set-up interview” related to the match with Bob. Alice fills in a form providing all the information needed to set up the interview, which are date and time of the meeting (04/07/2024 11:00 a.m.) and the link to a previously created virtual meeting on an external platform. After that, the system allows Alice to create a structured questionnaire to collect and evaluate Bob’s answers, as a complementary mechanism for the selection process. Alice creates the questionnaire by defining the questions for Bob, the possible answers and a different score for each answer; after that, she confirms its generation and also the set-up of the interview. As a consequence, the system shows the programmed meeting in the “Interview calendar” section of TF’s homepage, by attaching to the corresponding date the link for the meeting, the time and some of Bob’s personal information. Lastly, the system notifies Bob of the programmed interview and of the available questionnaire.

Student fills out a questionnaire for the selection process

Bob receives a notification by the system informing him that company TF has set-up an interview and created a questionnaire for the selection process of Int1. Bob logs in to the platform with his credentials and enters the “Interview calendar” section located in his homepage, where he can see a new programmed meeting for 04/07/2024 at 11:00 a.m. and the related questionnaire; therefore, Bob fills out the questionnaire and submits it. As a consequence, the system evaluates Bob’s answers with a grade and updates the suitability rank in the “Completed matches” section for Int1 in TF’s homepage with the calculated grade; the system also notifies company TF about this action.

Company admits a student for an internship

After interviewing all the matching students for Int1, company TF wants to admit those of them who look more suitable. Alice enters the section dedicated to Int1 and goes to the subsection “Completed matches”. Based on the high position of Bob in the suitability rank and on the positive outcome of his interview, Alice decides to admit him by clicking the button “Admit” associated to Bob in the rank. As a consequence, the system notifies Bob about his admission and adds in the section “Active internships” of his homepage an entry dedicated to the new ongoing internship, with information about the date of enrollment, some of TF’s identification data and the description document of Int1. Vice versa, Alice enters the subsection “Enrolled students” of section “Int1” and can now view a new line dedicated to Bob’s ongoing internship, with attached information such as the date of enrollment and Bob’s CV.

Users fill out the satisfaction questionnaires regarding an internship

After ending his internship project with company TF, Bob logs in to the platform with his credentials, enters the section “Satisfaction questionnaires” and opens the questionnaire related to Int1, where he expresses his positive opinion on the pathway by answering the specific questions generated by the system. In a second moment, Alice repeats the same process by filling out and submitting the satisfaction questionnaire related to the internship experience with Bob. After both questionnaires have been submitted, the system evaluates positively the progress of Int1, in order to feed statistical analysis for future recommendation processes.

Student communicates with a company

Bob is currently working on a project included in Int1 and is encountering difficulties in understanding a task specification in the internship document. Therefore, Bob logs in to

the S&C website with his credentials and enters the “Messaging” section of his homepage, where for each active internship the system provides a mechanism to communicate with the corresponding company. Bob opens the messaging session with company TF related to Int1 and complains about ambiguities in the description of that specific task; the system then notifies the company about the student’s message. After receiving the message, Bob’s tutor Carl logs in to the platform with the company’s credentials, enters the section dedicated to Int1 and goes into the “Messaging” subsection, where for each student carrying out the internship the system provides a mechanism to communicate with them. Carl reads Bob’s complaint and answers to his message by clarifying the task’s specification. Bob is then notified about the company’s response, so he reads the clarification (with the same described process) and can continue to work on the project.

2.1.2. Domain Class Diagram

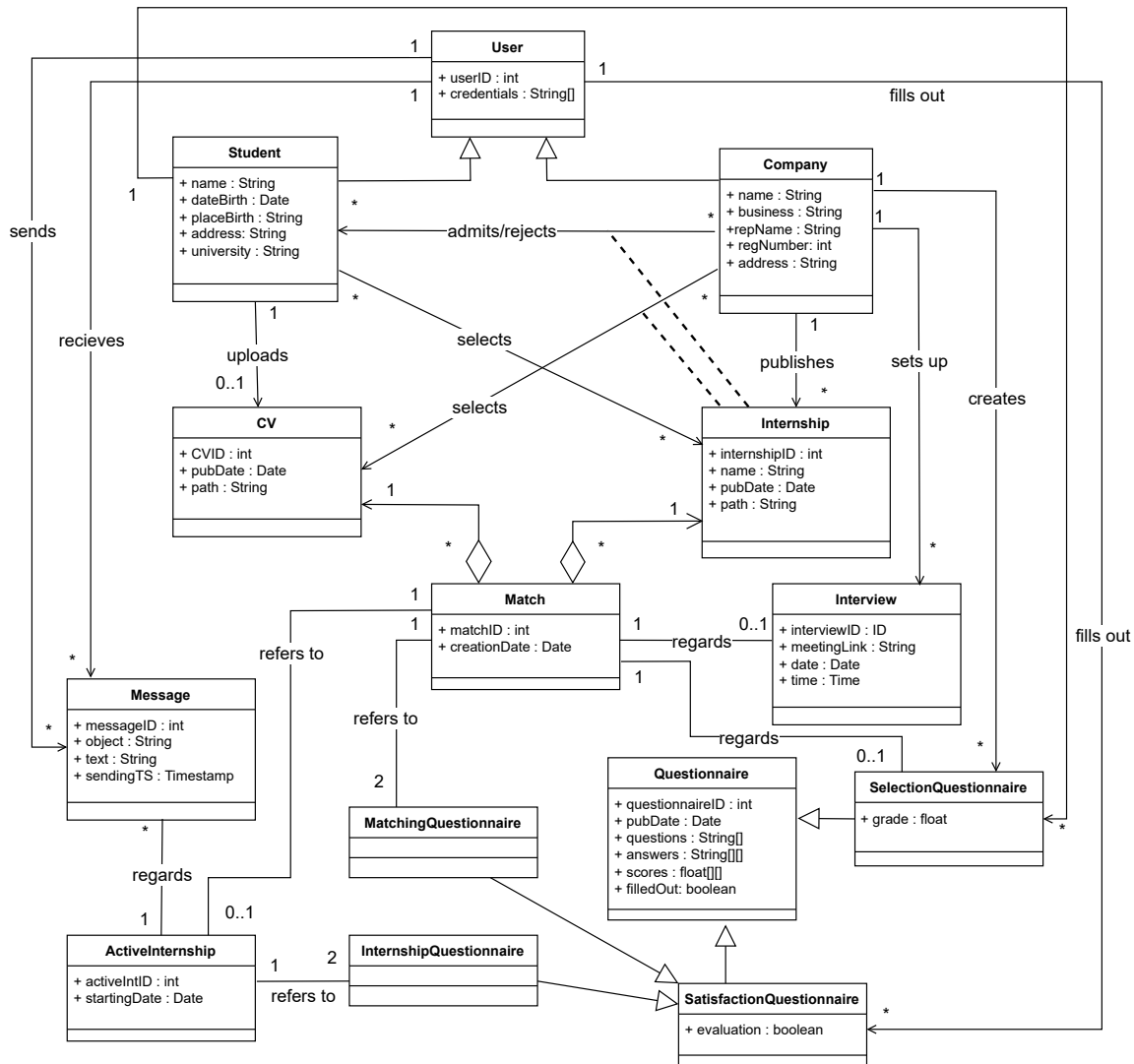


Figure 2.1: Domain Class Diagram

In Figure 2.1 you can find the domain class diagram of the application. As mentioned in the previous sections, the two main users of the system are students and companies. Students can select zero or many internships of their interest, while companies can select zero or many CVs for each of their offers. A match is created when a student and a company mutually select each other within the scope of an internship, therefore it associates a student CV with a specific internship. Companies can set up multiple interviews, one for each of their completed matches, and can do the same for the structured questionnaires that help them in the selection process (SelectionQuestionnaires). Symmetrically, students can fill out different questionnaires of this kind, one for each of their completed

matches. An active internship is created every time that a company admits one of their matching students for an internship. In this context, users can send/receive zero or more messages to/from the other party. Lastly, for each completed match and for each active internship, exactly two satisfaction questionnaires are available to be filled out by the involved student and company.

2.1.3. State Diagrams

Student process state diagram

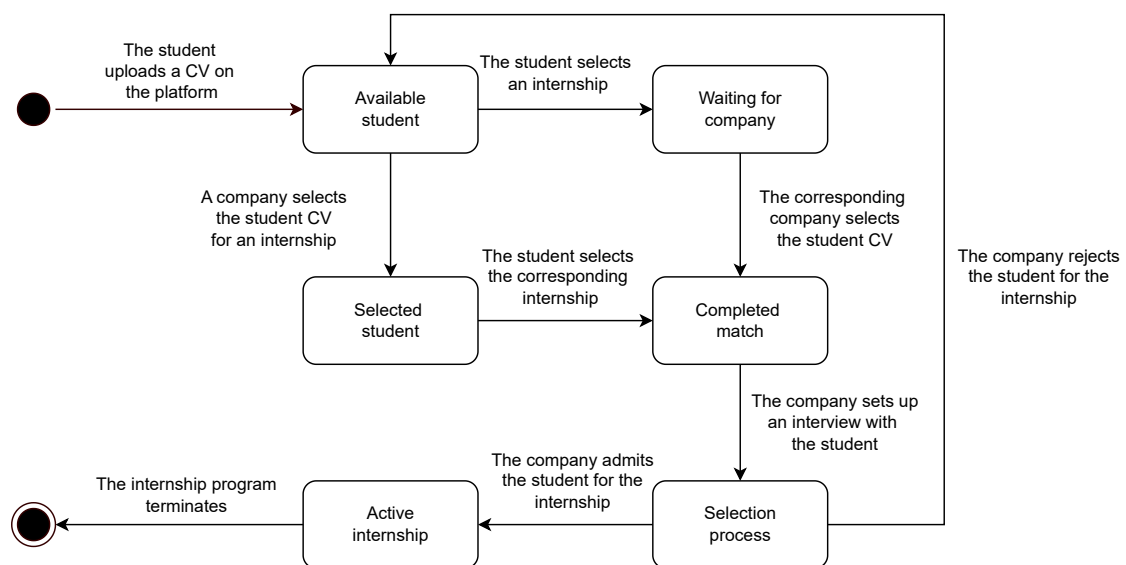


Figure 2.2: Student process state diagram

In Figure 2.2, the general process involving a registered student is shown, starting from the CV upload and ending with the carrying out of an internship. Note that the diagram describes just one matchmaking process, while a student can actually select/carry out multiple internships and create more matches simultaneously.

Company process state diagram

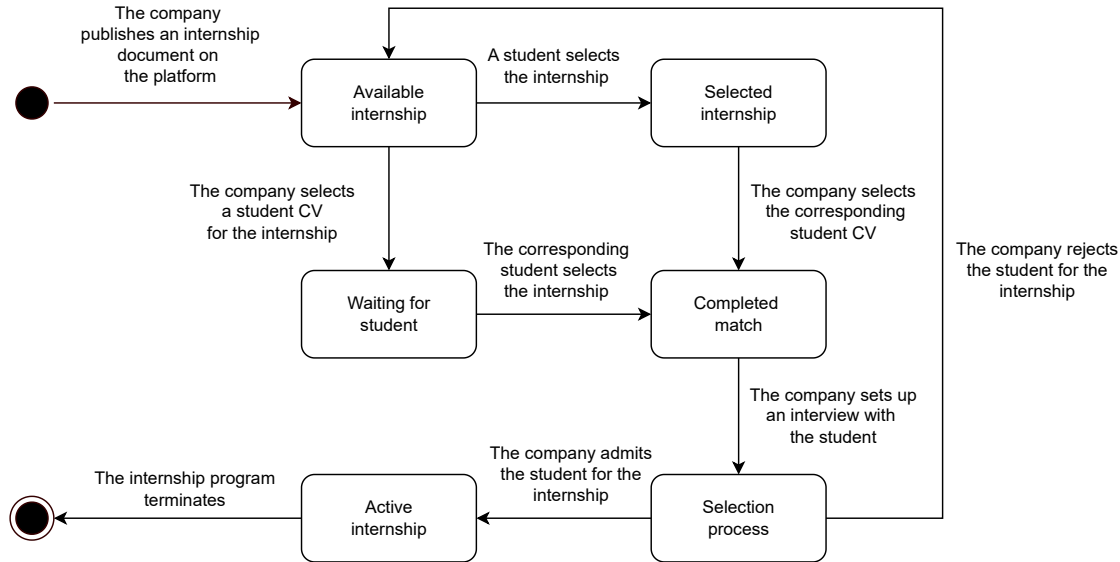


Figure 2.3: Company process state diagram

In Figure 2.3, the general process involving a registered company is shown, starting from the internship publication and ending with its activation for a specific student. Note that the diagram describes just one matchmaking process, while a company can actually publish different internships, select multiple CVs and create more matches simultaneously.

Satisfaction questionnaire state diagram

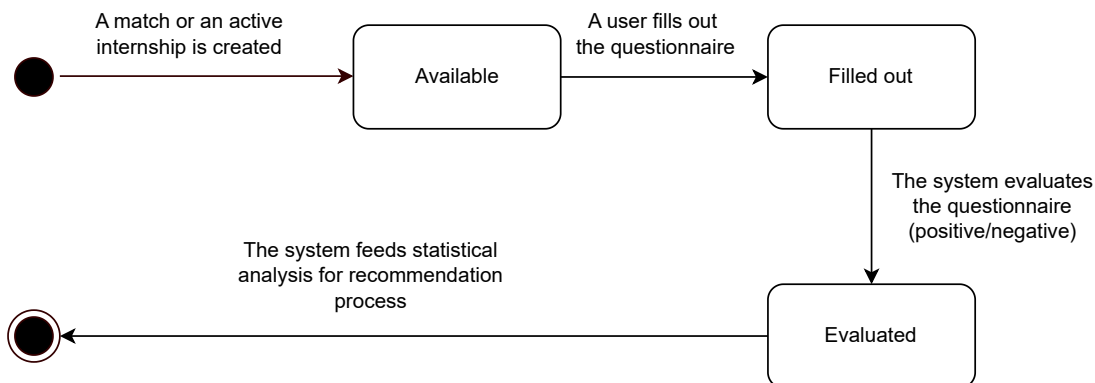


Figure 2.4: Satisfaction questionnaire state diagram

In Figure 2.4, the states of a satisfaction questionnaire about a match or an internship are shown. The system automatically provides users with it, and evaluates it once it has been filled out by a student or a company.

Selection process questionnaire state diagram

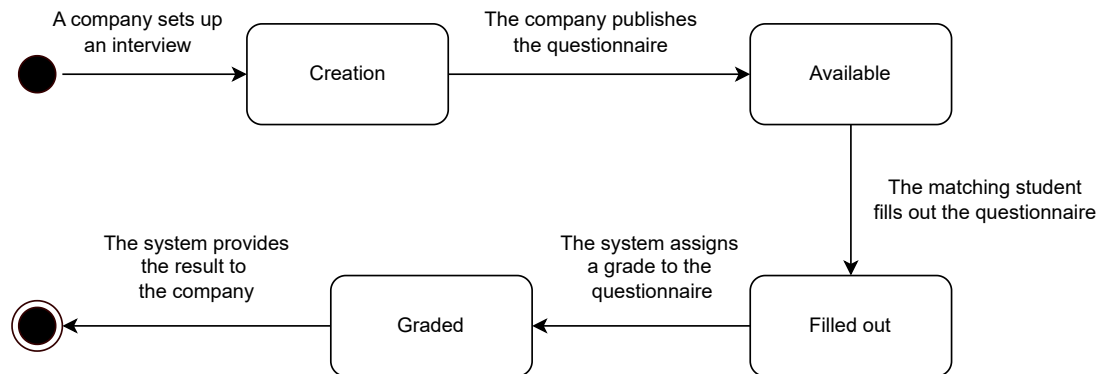


Figure 2.5: Selection process questionnaire state diagram

In Figure 2.5, the states of a selection process questionnaire are shown. A company can create and publish it after setting up an interview. Once the questionnaire has been filled out by a matching student, it is automatically graded by the system.

2.2. Product Functions

Sign up, log in, log out

When registering to the platform, users can decide to do that either as students or as companies, providing different kind of information, including their new credentials (email address and password). Students then have to authenticate (through an external platform) with their institutional credentials, while company identity is checked by the system thanks to a specific API. Once the system has checked that users with the same unique fields do not exist (such as email, registration number for companies, etc.), provided information is saved and the new user is from now on able to log in to the platform through their credentials. At any time when logged in, users are able to log out from the platform as well.

CV upload and internship publication

Once successfully registered and logged in, students can upload their CV, as a document containing their skills, attitudes and experiences, from the designated section. At any time, students can also remove their CV, and possibly substitute it with an updated version. Symmetrically, companies can publish their internships, as documents containing a detailed description of related projects and terms, from the designated section. Just like students with CVs, companies can remove their internship documents at any time

and replace them.

Matchmaking process

During this phase, both students and companies can initiate the process by selecting either an internship (for students) or a CV for an internship (for companies) among the available (and possibly recommended) ones, showing interest for the other part which is notified from the system. Once that happened, the notified actor can show mutual interest accepting the internship (students) or the CV (companies). If that happens, the other party is notified and a match is created. The system now enables the selection process tools for the company.

Selection process

During this phase, companies can decide which matching students to admit to their internships through the help of the system. They can set up interviews held on external platforms (such as Google Meet, Zooms, etc.), providing scheduling information, and create structured questionnaires to be filled out by students. Both students and companies are notified of other party's actions during this process, and at the end of the latter, companies decide whether to admit or reject students basing on their impression of them and on the grades of the questionnaires, which are automatically computed by the system. In case of admission, the internship is activated.

Student-Company communication

Once an internship is activated, both the involved student and company can provide information to the other party about the status of the ongoing internship, such as general feedback, suggestions or complaints. This is allowed by the system through a designated section, where both students and companies can also see what the other party has written.

Recommendation

This functionality allows both students and companies to look among internships and CVs that could interest them or correspond to their needs more than others. To achieve this, the system is able to recommend internships and CVs through different mechanisms. The first one is a straightforward keyword matching between such documents. On the other hand, the second one includes also statistical analysis based on satisfaction questionnaires provided to both students and companies about completed matches and active internships. The system is able to give a Boolean evaluation (positive or negative) to such questionnaires, and at the same time to associate similar students and internships

(through keyword matching between their CVs and documents). In this way, the system can recommend to students internships similar to the ones positively evaluated by similar students. Symmetrically, recommended CVs for companies are determined following the same logic.

2.3. User Characteristics

As the name tells, the two main classes of users of S&C are students and companies. The distinction between them is achieved during the registration to the platform.

Students are the ones looking for internships to carry out: after uploading a CV, they can initiate the matchmaking process by selecting an internship, or wait for a company to select their CV. They are the ones who undergo the selection process, and, as well as companies, they can communicate with them once internships are active.

On the other hand, **companies** are the ones looking for students to carry out their internships: after publishing one, they can initiate the matchmaking process by selecting a CV, or wait for students to show interest in their internships. The selection process is meant for them to admit suitable students.

Additionally, both students and companies are questioned by the system to feed statistical analysis for future recommendations through satisfaction questionnaires.

2.4. Assumptions, Dependencies and Constraints

S&C is meant to be a **web-based application**, accessible via web browsers from desktop and mobile devices.

Additionally, S&C does not support direct **interview** conduction, which is **externally managed** by companies using a virtual meeting platform (e.g. Google Meet, Zoom, etc.).

Finally, S&C will process user information in line with **GDPR** (General Data Protection Regulation) standards, to ensure data protection and privacy.

2.4.1. Domain Assumptions

The following assumptions are facts that the system cannot check and has to take for granted in order to ensure the correct and expected behavior of the platform.

[D1] Students' information contained in their published CVs must be correct.

[D2] Students must remove their uploaded CVs if these are obsolete or contain incorrect information.

[D3] For each company, internships' information (such as projects and terms) contained in the related published documents must be correct.

[D4] Companies must remove their published internships' documents if these are obsolete or if the related internships are no longer available.

[D5] Users must have a reliable internet connection.

[D6] Companies must be able to create a virtual meeting for an interview and to provide the related link to the system.

[D7] For each of their active internships, users must fill out the satisfaction questionnaire provided by the system with correct and coherent information.

[D8] For each of their completed matchmaking processes, users must fill out the satisfaction questionnaire provided by the system with correct and coherent information.

[D9] APIs and related platform used for external student authentication and company identity verification must work properly.

3 | Specific Requirements

3.1. External Interface Requirements

3.1.1. User Interfaces

This section provides the mockups of the main user interfaces, to be used as a guide during the implementation phase.

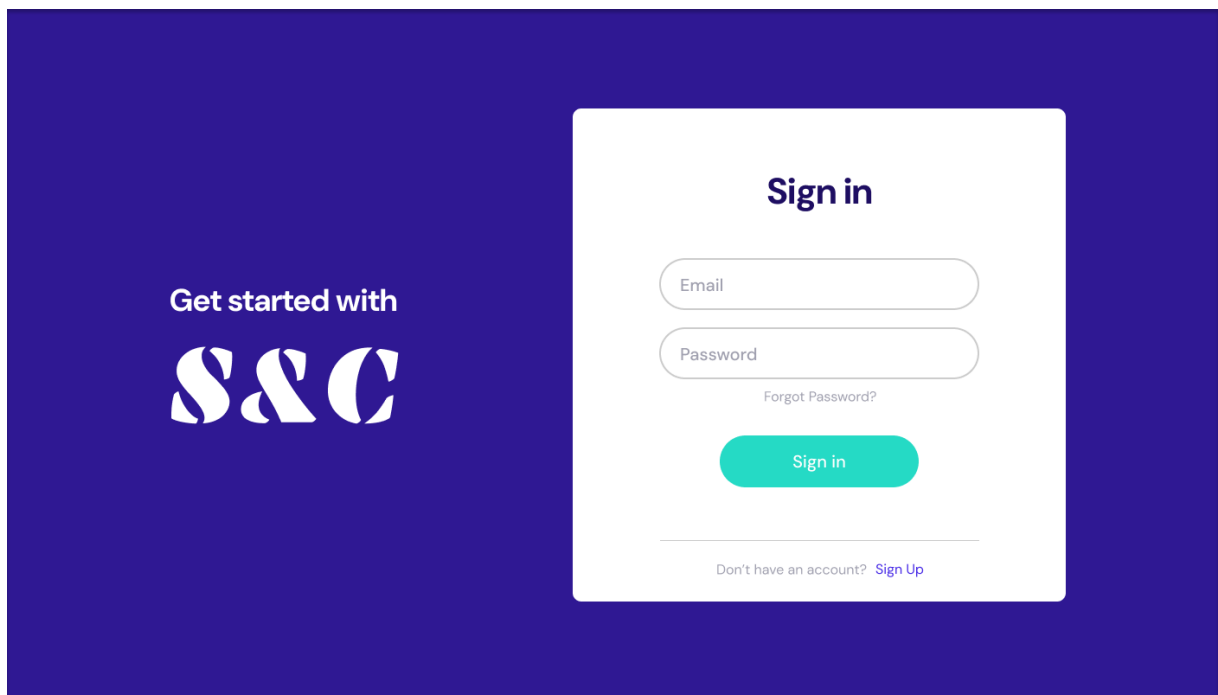
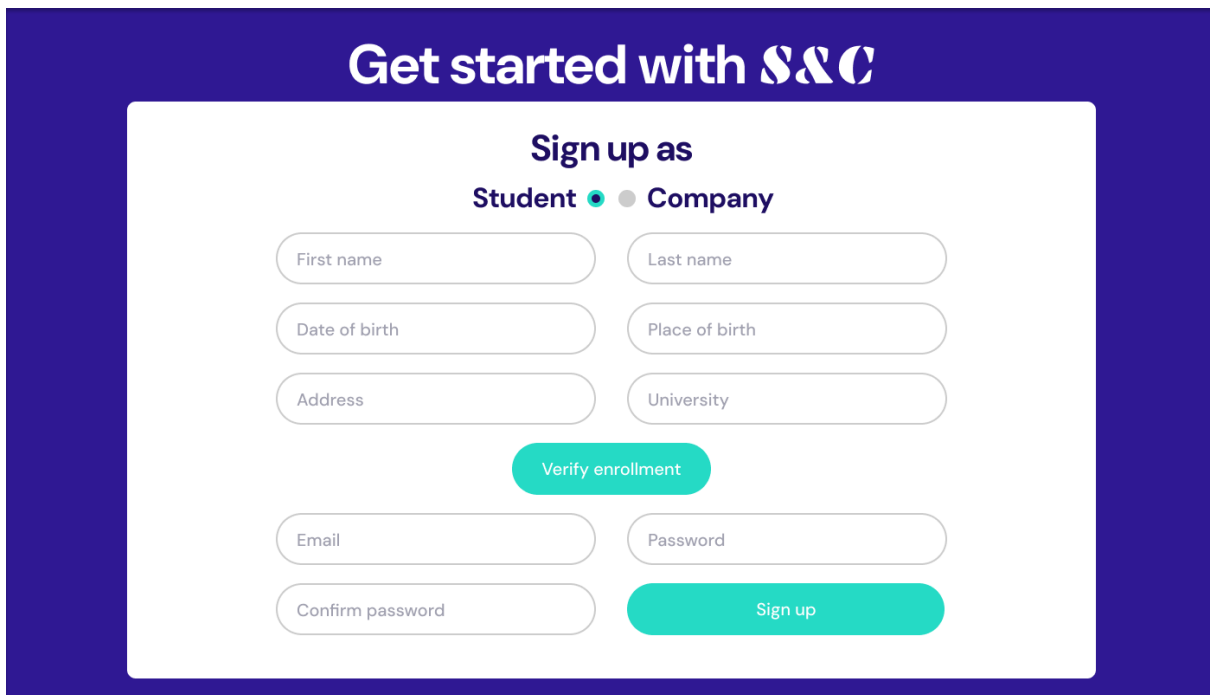


Figure 3.1: Login page UI

The login page, common to both students and companies, is shown in Figure 3.1.

Figures 3.2 and 3.3 show the registration interfaces of students and companies. By clicking on "Verify enrollment", students are redirected to an external platform where they can authenticate using their institutional credentials.



The image shows a registration form titled "Get started with S&C" on a dark blue background. The form is titled "Sign up as" and has two radio buttons: "Student" (selected) and "Company". The form fields are arranged in two columns. The left column contains "First name", "Date of birth", "Address", "Email", and "Confirm password". The right column contains "Last name", "Place of birth", "University", "Password", and a "Sign up" button. A "Verify enrollment" button is located between the two columns, below the "Address" and "University" fields.

Get started with S&C

Sign up as

Student ☒ Company ☐

First name Last name

Date of birth Place of birth

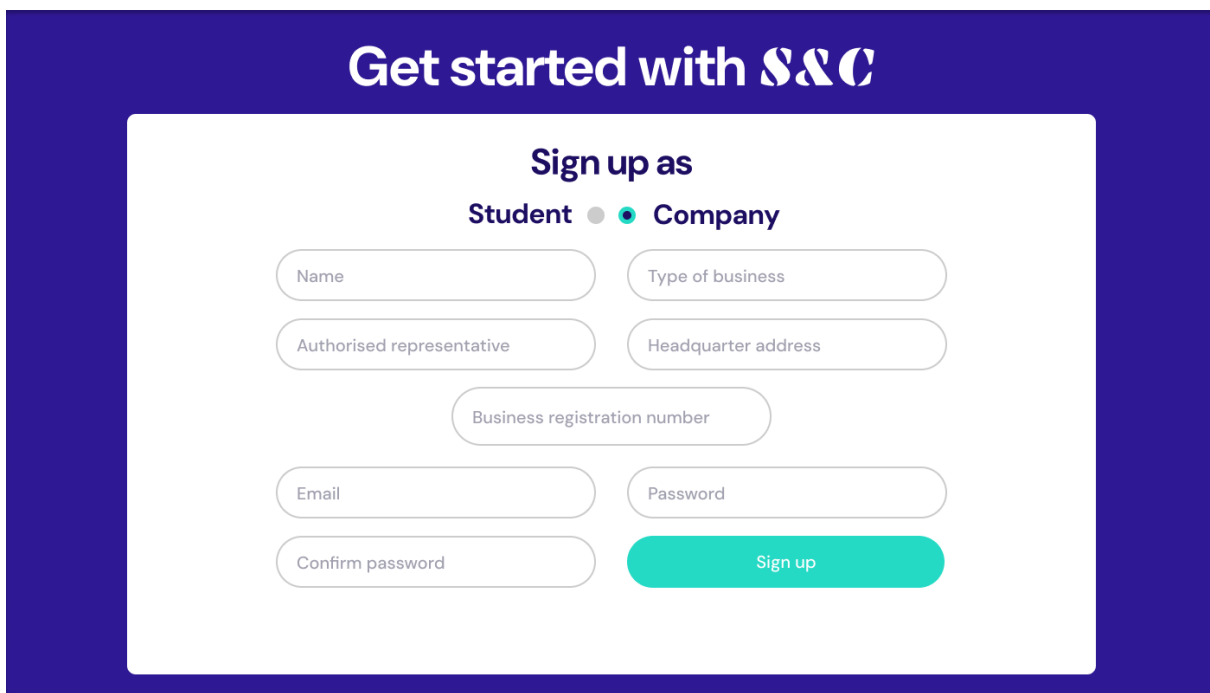
Address University

Verify enrollment

Email Password

Confirm password Sign up

Figure 3.2: Student registration UI



The image shows a registration form titled "Get started with S&C" on a dark blue background. The form is titled "Sign up as" and has two radio buttons: "Student" (unselected) and "Company" (selected). The form fields are arranged in two columns. The left column contains "Name", "Authorised representative", "Email", and "Confirm password". The right column contains "Type of business", "Headquarter address", "Business registration number", "Password", and a "Sign up" button.

Get started with S&C

Sign up as

Student ☐ Company ☒

Name Type of business

Authorised representative Headquarter address

Business registration number

Email Password

Confirm password Sign up

Figure 3.3: Company registration UI

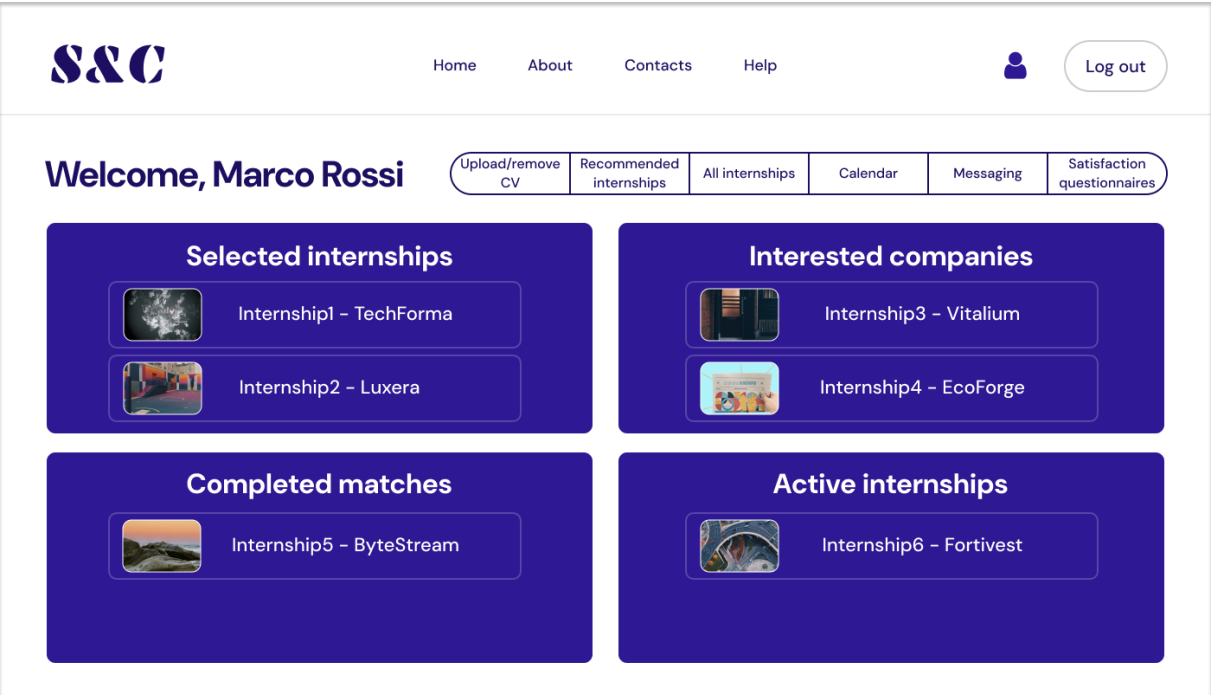


Figure 3.4: Student homepage UI

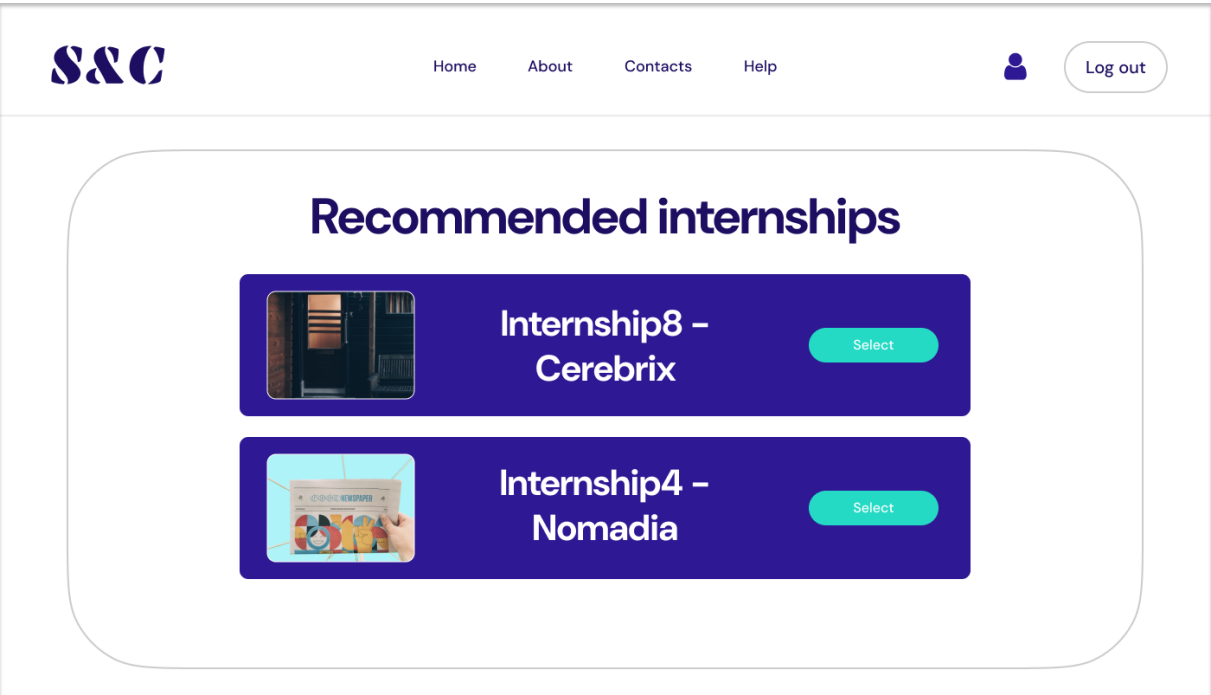


Figure 3.5: Student recommended internships UI

In Figure 3.4 students' homepage interface is shown. It provides an overview of the main sections they can access.

Figure 3.5 shows the recommended internships section for students. Note that, even if not reported in this document, a totally analogous interface for companies can be considered with recommended students for each of their internships.

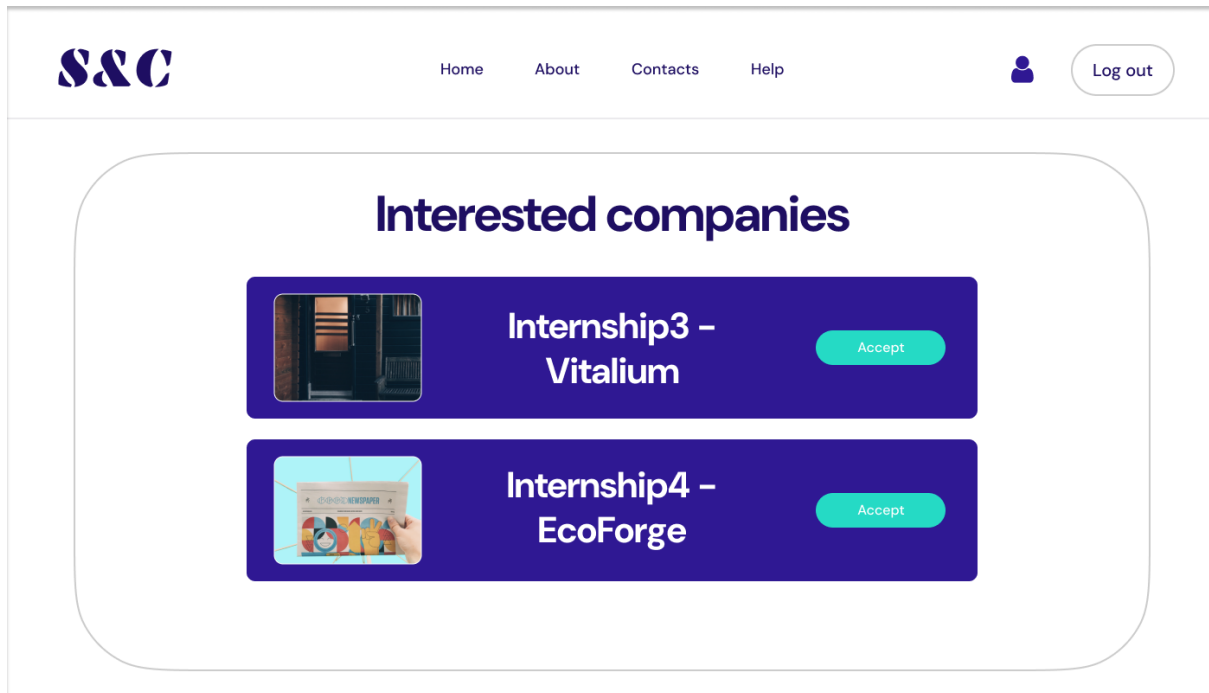


Figure 3.6: Student interested companies UI

Figure 3.6 shows the interested companies section for students. Note that, even if not reported in this document, a totally analogous interface for companies can be considered with interested students for each of their internships.

In Figure 3.7 the students' interview calendar is shown, just after clicking on a specific interview. Note that, even if not reported in this document, a totally analogous interface can be considered for companies, simply without the "Fill out questionnaire" option.

Figure 3.8 shows the students' messaging section, containing a chat for each of their active internships. Note that, even if not reported in this document, a totally analogous interface for each internship can be considered for companies, with a chat for every enrolled student.

Figure 3.9 shows the student interface for the satisfaction questionnaires section. Note that, even if not reported in this document, a totally analogous interface for each internship can be considered for companies.

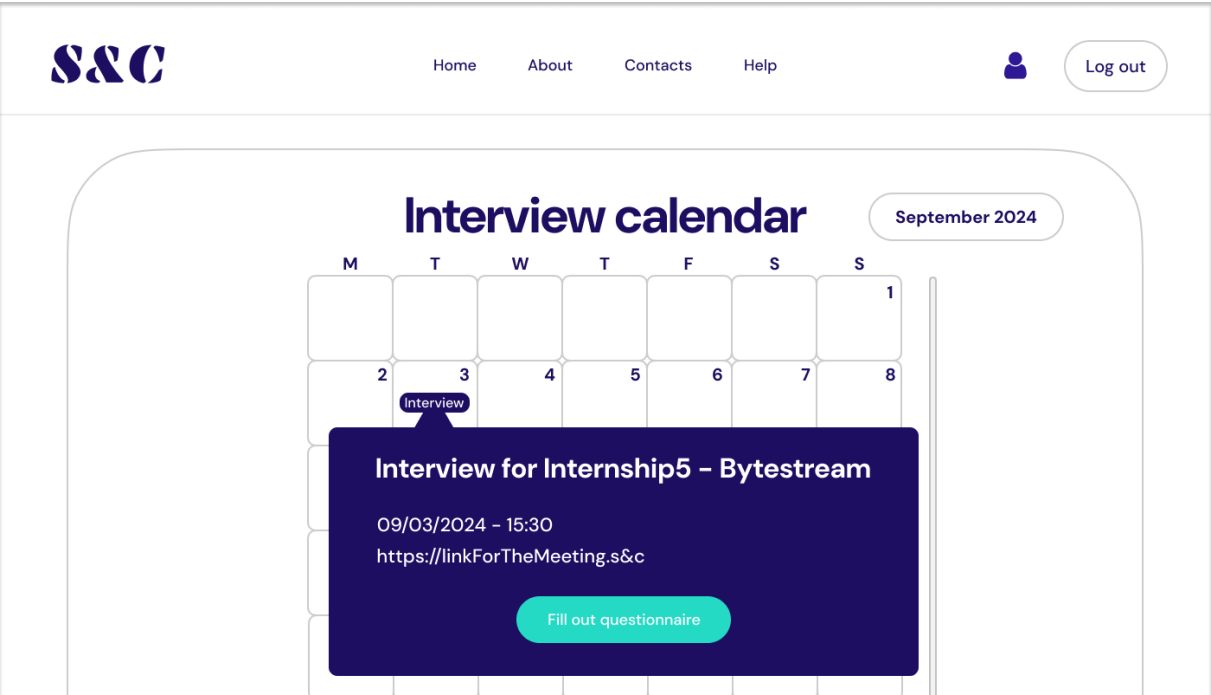


Figure 3.7: Student interview calendar UI

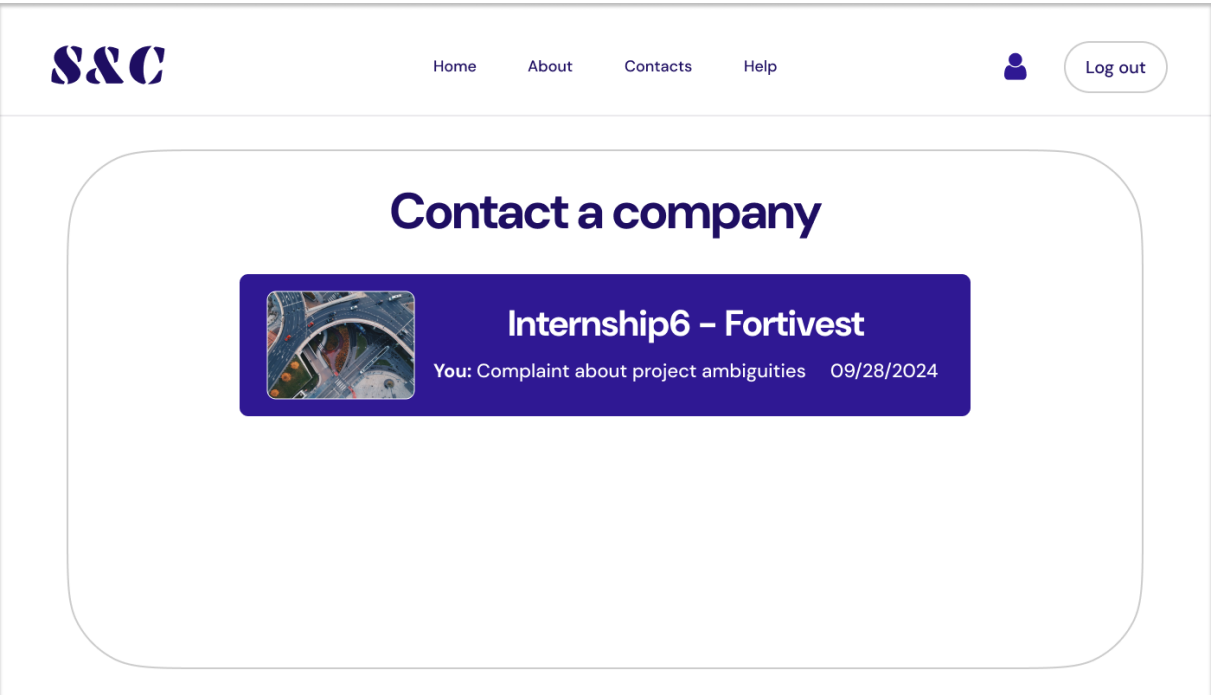


Figure 3.8: Student messaging section UI

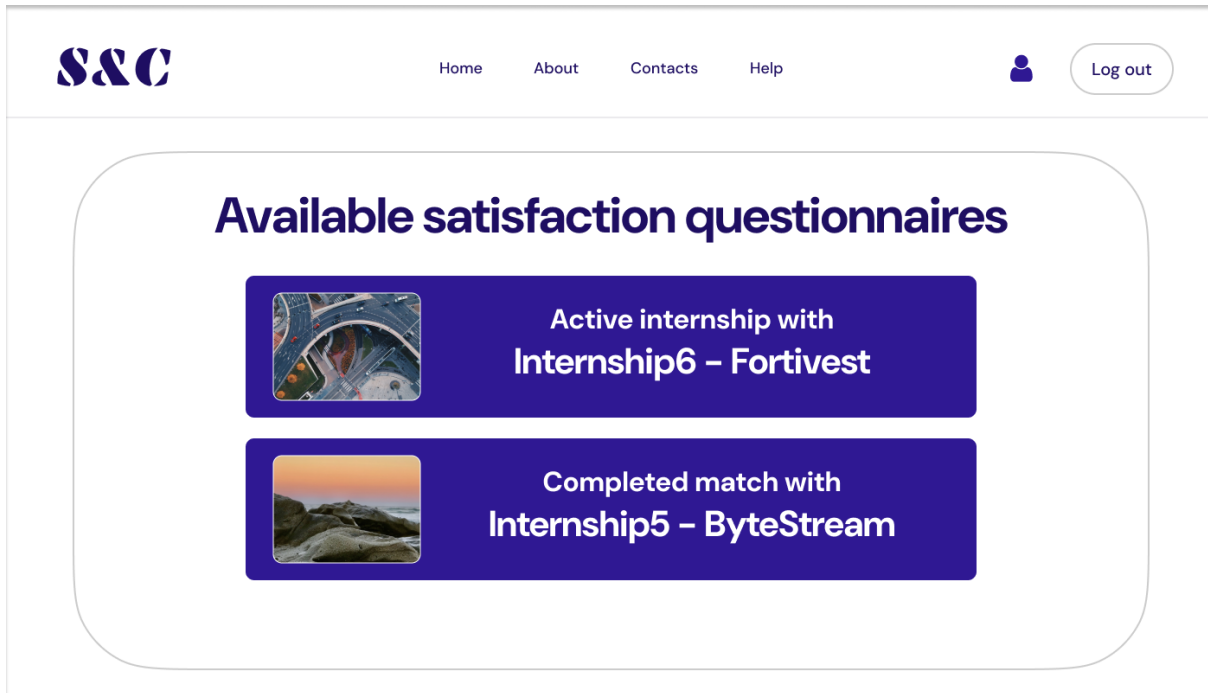


Figure 3.9: Student satisfaction questionnaires section UI

In Figure 3.10 the company homepage interface is shown. From there, companies can access a section dedicated to each of their internships.

Figure 3.11 shows the internship section interface for companies. It provides an overview of the main sections they can access.

Figure 3.12 shows the company interface dedicated to completed matches for an internship. From there, companies can set up interviews and admit/reject students based on the suitability rank.

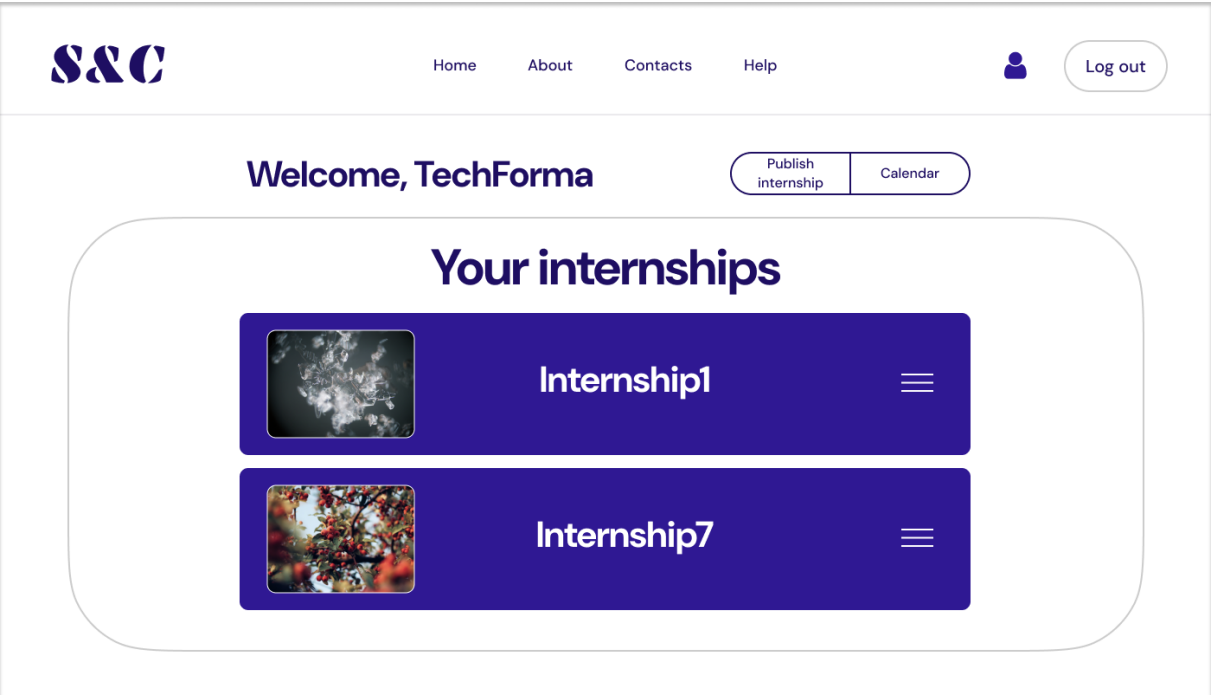


Figure 3.10: Company homepage UI

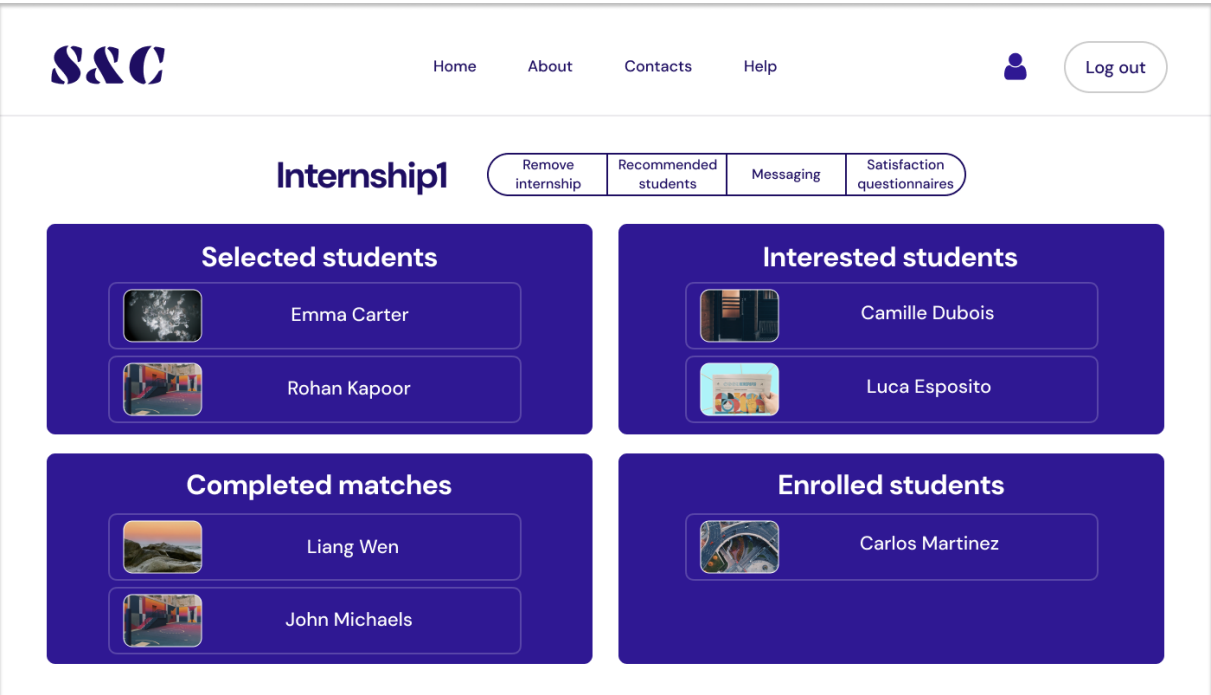


Figure 3.11: Company internship section UI

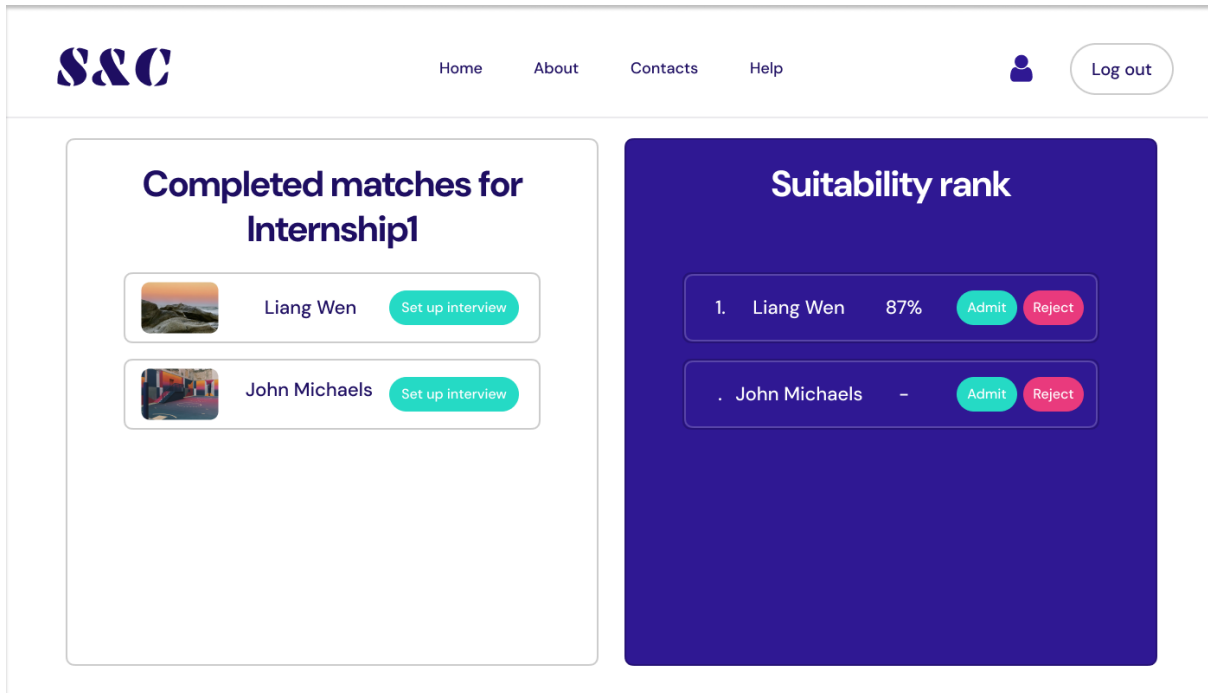


Figure 3.12: Company completed matches UI

3.1.2. Hardware Interfaces

As previously stated, S&C will be a web-application, not needing any specific hardware interface. To access the platform, everything users will need is a desktop or mobile device with a web browser and a functioning Internet connection.

3.1.3. Software Interfaces

S&C will need specific external APIs to correctly and legally authenticate university students and companies at the time of registration.

- **Student authentication API** (UNiDAYS): it is used to check whether a student is currently enrolled at a university. The student is redirected to UNiDAYS, where they can authenticate using institutional credentials, and eventually successfully redirected to S&C.
- **Company authentication API** (OpenCorporates API): it is used to directly check whether a company has provided valid information, such as name and registration number.

3.1.4. Communication Interfaces

For all client-server communication, the S&C application should utilize RESTful APIs, for a stateless, scalable and efficient interactions, based on HTTPS protocol and JSON data format.

3.2. Functional Requirements

Sign up, log in, log out

[R1] The system should allow new users (i.e. they are not registered yet) to register as students by providing their personal information, a valid email address and a password¹.

[R2] The system should allow new users (i.e. they are not registered yet) to register as companies by providing company identification information, a valid email address and a password².

[R3] The system should allow registered users to log in using their email address and password.

[R4] The system should allow all logged in users to log out from the platform.

Students

[R5] The system should allow students to upload their CV on the platform only if a CV of that student is not currently uploaded.

[R6] The system should allow students to remove their uploaded CV from the platform³.

[R7] The system should allow students to look among all available internships (and related document) published by companies on the platform.

[R8] The system should allow students to look among internships (and related document) that are recommended by the platform.

[R9] The system should allow students who have already uploaded their CV to select one or more internships of their interest among the available ones (i.e. students start the matching process showing interest for the internship).

[R10] The system should notify students whenever a new recommended internship has been published on the platform by a company.

¹See authentication requirement R54.

²See authentication requirement R55.

³The modification of a CV can be obtained by combining R5 and R6.

[R11] The system should notify students whenever a company, whose internship has been selected by that student (i.e. the student initiated the process), has accepted their CV for the selection process regarding that internship.

[R12] The system should notify students whenever a company selects their CV for an internship (i.e. the company initiates the process showing interest for a student CV).

[R13] The system should allow students to accept an internship whose company selected their CV for that internship (i.e. students end the matching process, the selection process starts).

[R14] The system should notify students whenever an interview for one of their matching internships has been programmed by a company.

[R15] The system should allow students to see their programmed interviews on a calendar.

[R16] The system should notify students whenever a questionnaire for one of their matching internships has been uploaded by a company.

[R17] The system should allow students to fill out available questionnaires for their matching internships.

[R18] The system should notify students when they are admitted to a matching internship after their interview with the company.

[R19] The system should notify students when they are rejected for a matching internship after their interview with the company.

[R20] The system should allow students to provide information to the other party about their active internships (such as complaints, problems or feedback).

[R21] The system should notify students whenever a company provides information (such as complaints, problems or feedback) about an active internship in which they are involved.

[R22] The system should notify students whenever a satisfaction questionnaire regarding an active internship or a matching process is available.

[R23] The system should allow students to fill out available satisfaction questionnaires.

Companies

[R24] The system should allow companies to publish their internships on the platform by providing a document containing information about projects and terms of the internship.

[R25] The system should allow companies to remove previously published internships from

the platform⁴.

[R26] For each of their internships, the system should allow companies to look among available students' CVs recommended by the platform.

[R27] For each of their internships, the system should allow companies to select one or more CVs among the recommended ones (i.e. the company initiates the matching process showing interest for a CV).

[R28] The system should notify companies whenever a new recommended CV for one of their internships has been uploaded by a student.

[R29] The system should notify companies whenever a student, whose CV has been selected by that company for a specific internship (i.e. the company initiated the matching process), has accepted that internship (i.e. the selection process starts).

[R30] The system should notify companies whenever a student selects one of their internships (i.e. the student initiates the matching process showing interest).

[R31] The system should allow companies to accept a CV of a student that selected one of their internships (i.e. companies end the matching process, starting the selection process).

[R32] For each of their internships, the system should allow companies to set up an interview with a matching student (for that internship), providing date, time and a web link for the meeting.

[R33] The system should allow companies to see their programmed interviews on a calendar.

[R34] For each of their internships, the system should allow companies to create structured questionnaires for the selection process of matching students.

[R35] The system should allow companies to upload created questionnaires, making them available for the matching students.

[R36] The system should notify companies whenever a matching student filled out the questionnaire for the selection process regarding one of their internships.

[R37] For each of their internships, the system should allow companies to see the results of the selection process questionnaires filled out by matching students ordered by grade.

[R38] For each of their internships, the system should allow companies to admit one or more interviewed matching students (i.e. the internship is now active).

⁴The modification of an internship can be obtained by combining R24 and R25.

[R39] For each of their internships, the system should allow companies to reject one or more interviewed matching students.

[R40] For each student carrying out an internship with them, the system should allow companies to provide information about that internship (such as complaints, problems or feedback).

[R41] For each of their active internships, the system should notify companies whenever a student involved in it provides information (such as complaints, problems or feedback).

[R42] The system should notify companies whenever a satisfaction questionnaire regarding an internship or a matching process is available.

[R43] The system should allow companies to fill out available satisfaction questionnaires.

System

[R44] The system should be able to evaluate with a grade selection process questionnaires filled out by students.

[R45] As soon as an internship is activated, the system should provide the involved student and company with a satisfaction questionnaire in order to collect statistical information about that internship (such as feedback and suggestions).

[R46] As soon as a student and company match, the system should provide both with a satisfaction questionnaire in order to collect information about the related matching process (such as feedback and suggestions).

[R47] The system should be able to recommend published internships to the students by matching keywords between their CVs and the internships' documents.

[R48] For each internship, the system should be able to recommend to the corresponding company available students' CVs by matching keywords between those and the internship document.

[R49] The system should be able to evaluate (positively or negatively) a completed match between a student and an internship offered by a company by analyzing the satisfaction questionnaires filled out by both parties.

[R50] The system should be able to associate students' CVs to similar ones involved in past completed matches (with internships) by keyword matching among CVs.

[R51] The system should be able to associate internships to similar ones involved in past completed matches (with students' CVs) by keyword matching among internships'

documents.

[R52] The system should be able to recommend to students internships similar (i.e. associated as in R51) to ones whose completed matching with similar (i.e. associated as in R50) students was positively evaluated.

[R53] For each internship, the system should be able to recommend to the corresponding company students' CVs similar (i.e. associated as in R50) to ones whose completed matching with similar (i.e. associated as in R51) internships was positively evaluated.

[R54] At the time of registration, the system should be able to validate student university enrollment by redirecting them to an external platform through an API, where they can authenticate using their institutional credentials.

[R55] At the time of registration, the system should be able to validate provided company information (such as registration number) through an external API.

3.2.1. Use Case Diagrams

In Figures 3.13 and 3.14 use case diagrams of student and company are shown. To better distinguish *initiate* and *participate* relationships, two different diagrams were created. In addition, for the sake of readability, *include* and *extend* relationships were not included, since the vast majority of use cases represent a unique flow of operations. Detailed information about dependencies among use cases is provided in the following section.

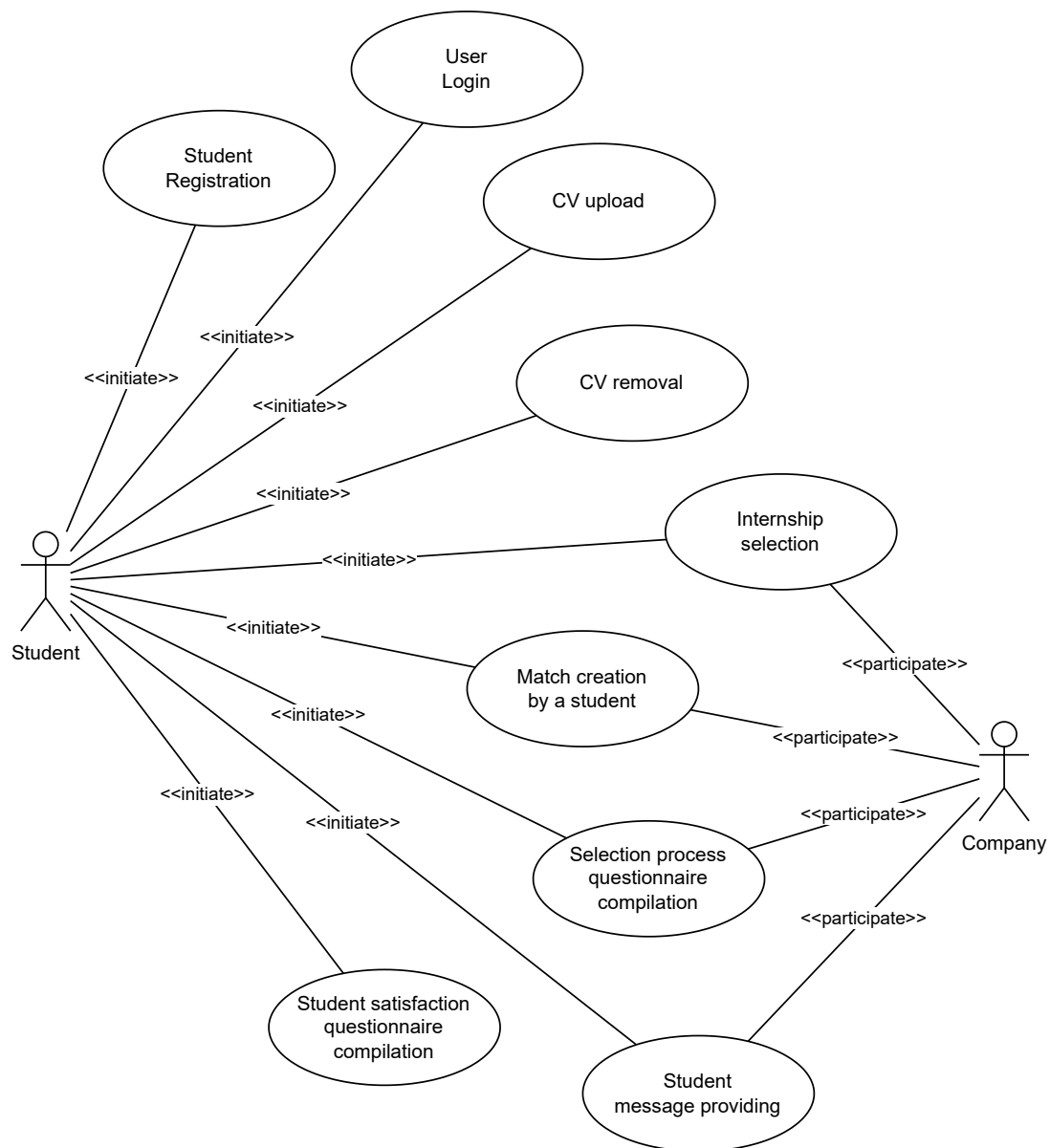


Figure 3.13: Student Use Case Diagram

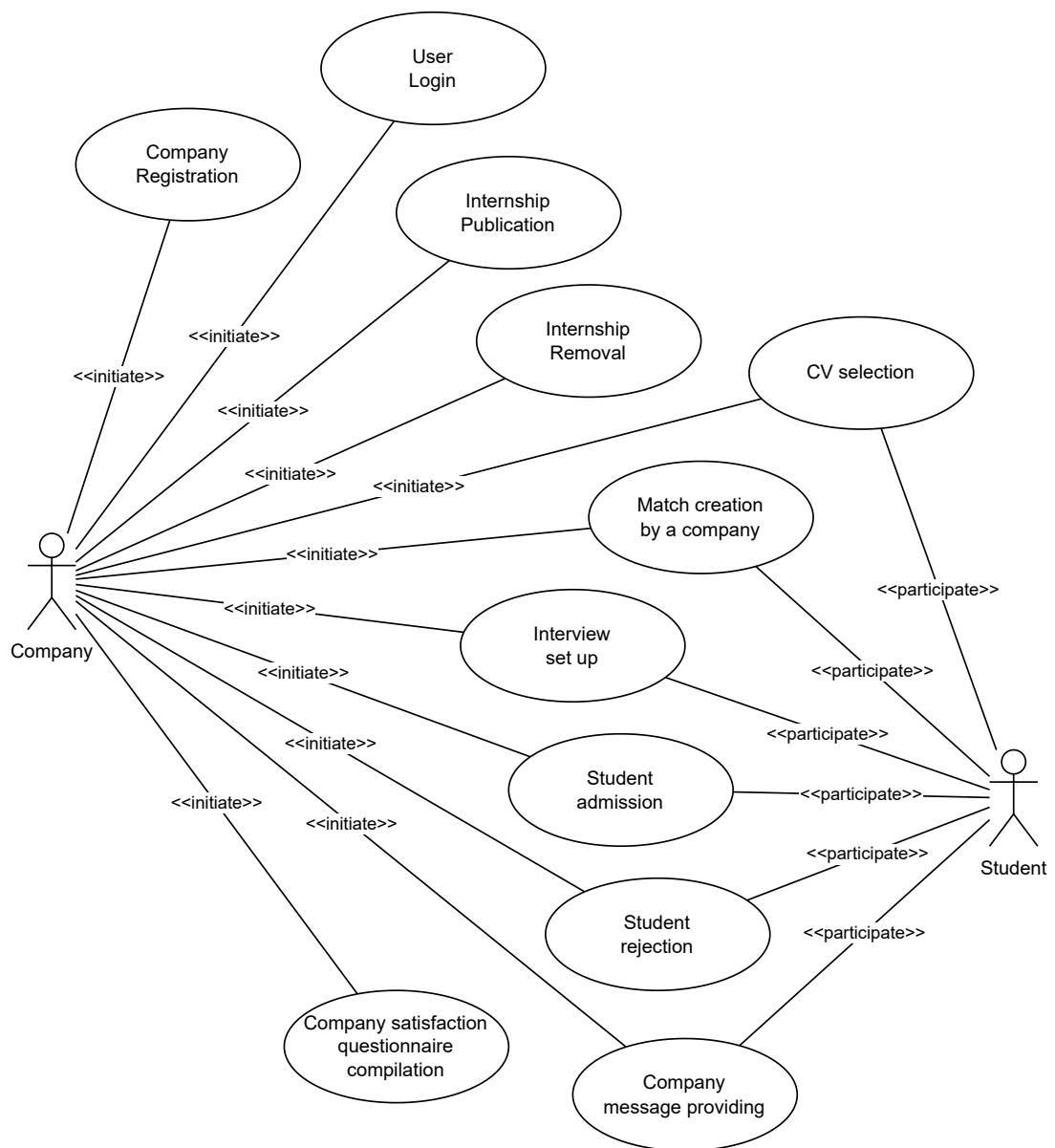


Figure 3.14: Company Use Case Diagram

3.2.2. Use Cases

[UC1] - Student registration

Name	Student registration
Actors	• Student
Entry condition	The student has opened the S&C website.
Event flow	1. The student clicks on “Register as student”. 2. The student inserts his personal information (complete name, gender, date and place of birth, address and name of the university in which he’s enrolled). 3. The student clicks on "Verify enrollment". 4. The system redirects the student to an external platform for university enrollment verification. 5. The student authenticates through their institutional credentials. 6. The system redirects the student back to the registration form. 7. The student inserts his personal e-mail and a password as their new credentials. 8. The student clicks on “Confirm registration”. 9. The system redirects the student to the Log-in page.
Exit condition	Student’s registration has been successfully executed.
Exceptions	5. The student is not enrolled in the specified university. The external platform will show an error message to the student and the system will redirect the student to the registration page. 8. There already exists a registered student with the inserted e-mail address. The system will show an error message to the student, asking to insert a new e-mail.

Table 3.1: UC1 - Student registration

[UC2] - Company registration

Name	Company registration
Actors	• Company
Entry condition	The company has opened the S&C website.
Event flow	1. The company clicks on “Register as company”. 2. The company inserts its identification information (name of the company, type of business, name of the authorized representative, business registration number and address of the headquarter). 3. The company inserts an e-mail and a password as its new credentials. 4. The company clicks on “Confirm registration”. 5. The system checks the existence of the company through an external platform. 6. The system redirects the company to the Log-in page.
Exit condition	Company’s registration has been successfully executed.
Exceptions	4. There already exists a registered company with that e-mail address. The system will show an error message to the company, asking to insert a new e-mail. 5. The specified business registration number doesn’t exist. The system will show an error message to the company.

Table 3.2: UC2 - Company registration

[UC3] - User login

Name	User login
Actors	• User
Entry condition	The user has opened the S&C website.
Event flow	1. The user clicks on “Login”. 2. The user inserts his e-mail and password. 3. The system checks user’s credentials. 4. The system redirects user to its S&C homepage.
Exit condition	User’s Login has been successfully executed.
Exceptions	3. The inserted e-mail or password is not correct (or both).

Table 3.3: UC3 - User login

[UC4] - CV upload

Name	CV upload
Actors	• Student
Entry condition	The student is logged in S&C and has not a currently uploaded CV.
Event flow	1. The student clicks on “Upload new CV”. 2. The student attaches his CV. 3. The student clicks on “Confirm CV upload”. 4. The system shows a message to the student of successful CV upload.
Exit condition	The student CV is available for selection and recommendation.
Exceptions	3. The format of the CV document is not supported.

Table 3.4: UC4 - CV upload

[UC5] - CV removal

Name	CV removal
Actors	• Student
Entry condition	The student is logged in S&C and has a currently uploaded CV.
Event flow	1. The student clicks on “Remove current CV”. 2. The student clicks on “Confirm CV removal”. 3. The system shows a message to the student of successful CV removal.
Exit condition	The student CV is no more available for selection and recommendation.
Exceptions	

Table 3.5: UC5 - CV removal

[UC6] - Internship publication

Name	Internship publication
Actors	• Company
Entry condition	The company is logged in S&C.
Event flow	1. The company clicks on “Publish new internship”. 2. The company inserts the internship title. 3. The company attaches the internship document. 4. The company clicks on “Confirm internship publication”. 5. The system creates a section dedicated to the new internship in the company’s homepage. 6. The system redirects the company to the internship section.
Exit condition	The internship is available for selection and recommendation.
Exceptions	4. The format of the document is not supported.

Table 3.6: UC6 - Internship publication

[UC7] - Internship removal

Name	Internship removal
Actors	• Company
Entry condition	The company is logged in S&C.
Event flow	1. The company enters the section dedicated to the internship to remove. 2. The company clicks on “Remove internship”. 3. The company clicks on “Confirm internship removal”. 4. The system shows a message to the company of successful internship removal.
Exit condition	The internship is no more available for selection and recommendation.
Exceptions	

Table 3.7: UC7 - Internship removal

[UC8] - CV selection

Name	CV selection
Actors	• Student • Company
Entry condition	The company is logged in S&C.
Event flow	1. The company enters a specific internship section. 2. The company enters the section “Recommended students”. 3. The system shows a list of recommended CVs. 4. The company clicks on a CV. 5. The system shows the content of the CV. 6. The company clicks on “Select student CV”. 7. The system adds the CV in the subsection “Selected students” of the internship section. 8. The system adds the corresponding internship document in the section “Interested companies” of the student’s homepage. 9. The system shows a message to the company of successful CV selection. 10. The system notifies the corresponding student.
Exit condition	The student is able to accept the internship.
Exceptions	3. There are no recommended CVs in that specific moment. The system will warn the company.

Table 3.8: UC8 - CV selection

[UC9] - Internship selection

Name	Internship selection
Actors	• Student • Company
Entry condition	The student is logged in S&C and has a currently uploaded CV.
Event flow	1. The student enters the section “Recommended Internships”. 2. The system shows a list of recommended internship documents. 3. The student clicks on an internship. 4. The system shows the content of the document. 5. The student clicks on “Select internship”. 6. The system adds the internship document in the section “Selected internships” of the student’s homepage. 7. The system adds the student’s CV in the subsection “Interested students” of the section dedicated to the internship, in the company’s homepage. 8. The system shows a message to the student of successful internship selection. 9. The system notifies the corresponding company.
Exit condition	The company is able to accept the student CV for the internship.
Exceptions	2. There are no recommended internships in that specific moment. The system will warn the student.

Table 3.9: UC9 - Internship selection

[UC10] - Match creation by a student

Name	Match creation by a student
Actors	• Student • Company
Entry condition	The student is logged in S&C and a company has selected their CV for an internship.
Event flow	1. The student enters the section “Interested companies”. 2. The system shows a list of internship documents for which companies have selected the student’s CV. 3. The student clicks on an internship. 4. The system shows the content of the document. 5. The student clicks on “Accept internship”. 6. The system adds the internship document in the section “Completed matches” of the student’s homepage. 7. The system adds the student’s CV in the subsection “Completed matches” of the section dedicated to the internship, in the corresponding company’s homepage. 8. The system shows a message to the student of successful match creation. 9. The system notifies the company. 10. The system provides the student with a satisfaction questionnaire for the matchmaking process in the section “Satisfaction questionnaires”. 11. The system provides the company with a satisfaction questionnaire for the matchmaking process in the subsection “Satisfaction questionnaires”. 12. The system notifies the student about the available questionnaire. 13. The system notifies the company about the available questionnaire.
Exit condition	The student has successfully concluded a matchmaking process and is available for the selection process.
Exceptions	

Table 3.10: UC10 - Match creation by a student

[UC11] - Match creation by a company

Name	Match creation by a company
Actors	• Company • Student
Entry condition	The company is logged in S&C and a student has selected one of its internships.
Event flow	1. The company enters a specific internship section. 2. The company enters the section “Interested students”. 3. The system shows a list of CVs associated to students that have selected the internship. 4. The company clicks on a CV. 5. The system shows the content of the CV. 6. The company clicks on “Accept student CV”. 7. The system adds the CV in the subsection “Completed matches” of the internship section. 8. The system adds the corresponding internship document in the section “Completed matches” of the student’s homepage. 9. The system shows a message to the company of successful match creation. 10. The system notifies the student. 11. The system provides the company with a satisfaction questionnaire for the matchmaking process in the subsection “Satisfaction questionnaires”. 12. The system provides the student with a satisfaction questionnaire for the matchmaking process in the section “Satisfaction questionnaires”. 13. The system notifies the company about the available questionnaire. 14. The system notifies the student about the available questionnaire.
Exit condition	The company has successfully concluded a matchmaking process with a student and is able to set up an interview for the selection process with them.
Exceptions	

Table 3.11: UC11 - Match creation by a company

[UC12] - Interview set up

Name	Interview set up
Actors	• Company • Student
Entry condition	The company is logged in S&C and is involved in at least one match with a student w.r.t. a specific internship.
Event flow	1. The company enters a specific internship section. 2. The company enters the section “Completed matches”. 3. The system shows a list of completed matches for the internship. 4. The company clicks on a match. 5. The system shows the CV of the matching student and the match creation date. 6. The company clicks on “Set up interview”. 7. The system shows the company a form to compile. 8. The company compiles the form by inserting date and time of the meeting and a link to an external meeting platform. 9. The system shows the company a form to create a structured questionnaire for the selection process. 10. The company compiles the form. 11. The company clicks on “Confirm interview set up”. 12. The system adds the programmed interview (date, time and link) in the section “Interview Calendar” of the company’s homepage. 13. The system adds the programmed interview and the questionnaire in the section “Interview calendar” of the student’s homepage. 14. The system notifies the student about the programmed interview. 15. The system notifies the student about the available questionnaire.
Exit condition	The company has successfully set up an interview with a matching student, who can now fill out the questionnaire for the selection process.
Exceptions	8. The company has already programmed an interview for that date and at that time. The system will warn the company and will ask for new information.

Table 3.12: UC12 - Interview set up

[UC13] - Selection process questionnaire compilation

Name	Selection process questionnaire compilation
Actors	• Company • Student
Entry condition	The student is logged in S&C and has an available selection process questionnaire to fill out.
Event flow	1. The student enters the section “Interview calendar”. 2. The system shows the student a calendar with his programmed interviews. 3. The student clicks on one of the interviews. 4. The system shows the student the date and time of the interview, the related internship document, the link for the meeting and the questionnaire for the selection process. 5. The student clicks on the questionnaire. 6. The system shows the student a form to compile. 7. The student fills out the questionnaire. 8. The student clicks on “Submit”. 9. The system grades the questionnaire. 10. The system updates the students’ suitability rank in the subsection “Matching students” of the section dedicated to the internship (in the company’s homepage), based on the student’s grade. 11. The system notifies the company.
Exit condition	The company can view the position of the student in the suitability rank of matching students for the internship.
Exceptions	5. The questionnaire for the interview has already been filled out. The system will warn the student.

Table 3.13: UC13 - Selection process questionnaire compilation

[UC14] - Student admission

Name	Student admission
Actors	• Company • Student
Entry condition	The company is logged in S&C and is involved in a match with a student for a specific internship.
Event flow	1. The company enters a specific internship section. 2. The company enters the section “Completed matches”. 3. The system shows the suitability rank of matching students based on their grades in the questionnaire. 4. The company clicks on a matching student. 5. The company clicks on “Admit to the internship”. 6. The system adds to the subsection “Enrolled students” of the internship section an entry dedicated to the activated internship. 7. The system adds to the section “Active internships” of the student’s homepage an entry dedicated to the new internship. 8. The system shows the company a message of successful internship activation. 9. The system notifies the student. 10. The system provides the company with a satisfaction questionnaire for the active internship in the subsection “Satisfaction questionnaires”. 11. The system provides the student with a satisfaction questionnaire for the active internship in the section “Satisfaction questionnaires”. 12. The system notifies the company about the available questionnaire. 13. The system notifies the student about the available questionnaire.
Exit condition	The company has successfully activated an internship with a student.
Exceptions	

Table 3.14: UC14 - Student admission

[UC15] - Student rejection

Name	Student rejection
Actors	• Company • Student
Entry condition	The company is logged in S&C and is involved in a match with a student for a specific internship.
Event flow	1. The company enters a specific internship section. 2. The company enters the section “Completed matches”. 3. The system shows the suitability rank of matching students based on their grades in the questionnaire. 4. The company clicks on a matching student. 5. The company clicks on “Reject for the internship”. 6. The system notifies the student.
Exit condition	The company has successfully rejected a student for an internship.
Exceptions	

Table 3.15: UC15 - Student rejection

[UC16] - Student satisfaction questionnaire compilation

Name	Student satisfaction questionnaire compilation
Actors	• Student
Entry condition	The student is logged in S&C and has an available satisfaction questionnaire.
Event flow	1. The student enters the section “Satisfaction questionnaires”. 2. For each matchmaking process or active internship related to the student, the system shows a satisfaction questionnaire. 3. The student clicks on a questionnaire. 4. The system shows a form to fill out. 5. The student fills out the questionnaire. 6. The student clicks on “Submit”. 7. The system evaluates the questionnaire.
Exit condition	The system is able to feed statistical analysis for future recommendation processes.
Exceptions	

Table 3.16: UC16 - Student satisfaction questionnaire compilation

[UC17] - Company satisfaction questionnaire compilation

Name	Company satisfaction questionnaire compilation
Actors	• Company
Entry condition	The company is logged in S&C and has an available satisfaction questionnaire.
Event flow	1. The company enters a specific internship section. 2. The company enters the subsection “Satisfaction questionnaires”. 3. For each active internship and for each matchmaking process related to that internship, the system shows a satisfaction questionnaire. 4. The company clicks on a questionnaire. 5. The system shows a form to fill out. 6. The company fills out the questionnaire. 7. The company clicks on “Submit”. 8. The system evaluates the questionnaire.
Exit condition	The system is able to feed statistical analysis for future recommendation processes.
Exceptions	

Table 3.17: UC17 - Company satisfaction questionnaire compilation

[UC18] - Student message providing

Name	Student message providing
Actors	• Student • Company
Entry condition	The student is logged in S&C and is involved in an active internship.
Event flow	1. The student enters the section “Messaging”. 2. For each active internship carried out by the student, the system shows a messaging session with the corresponding company. 3. The student clicks on a messaging session. 4. The student sends a message to the corresponding company. 5. The system shows the new message in the current session. 6. The system shows the new message in the corresponding chat in the subsection “Messaging” of the section dedicated to the internship (in the company’s homepage). 7. The system notifies the company about the message.
Exit condition	The company has successfully received the message.
Exceptions	

Table 3.18: UC18 - Student message providing

[UC19] - Company message providing

Name	Company message providing
Actors	• Company • Student
Entry condition	The company is logged in S&C and is involved in an active internship.
Event flow	1. The company enters a specific internship section. 2. The company enters the subsection “Messaging”. 3. For each student actively enrolled in that internship, the system shows a messaging session with the student. 4. The company clicks on a messaging session. 5. The company sends a message to the corresponding student. 6. The system shows the new message in the current session. 7. The system shows the new message in the corresponding chat of the section “Messaging”, in the student’s homepage. 8. The system notifies the student about the message.
Exit condition	The student has successfully received the message.
Exceptions	

Table 3.19: UC19 - Company message providing**3.2.3. Sequence Diagrams**

In this section, the sequence diagrams describing the use cases of the previous section are shown. Except from the two main actors (student and company), the other entities that participate in the interactions are the following:

1. **System:** the S&C system.
2. **VerificationPlatform:** the student authentication platform (UNiDAYS).
3. **CompanyVerifier:** the company authentication platform (OpenCorporates).

[UC1] - Student registration

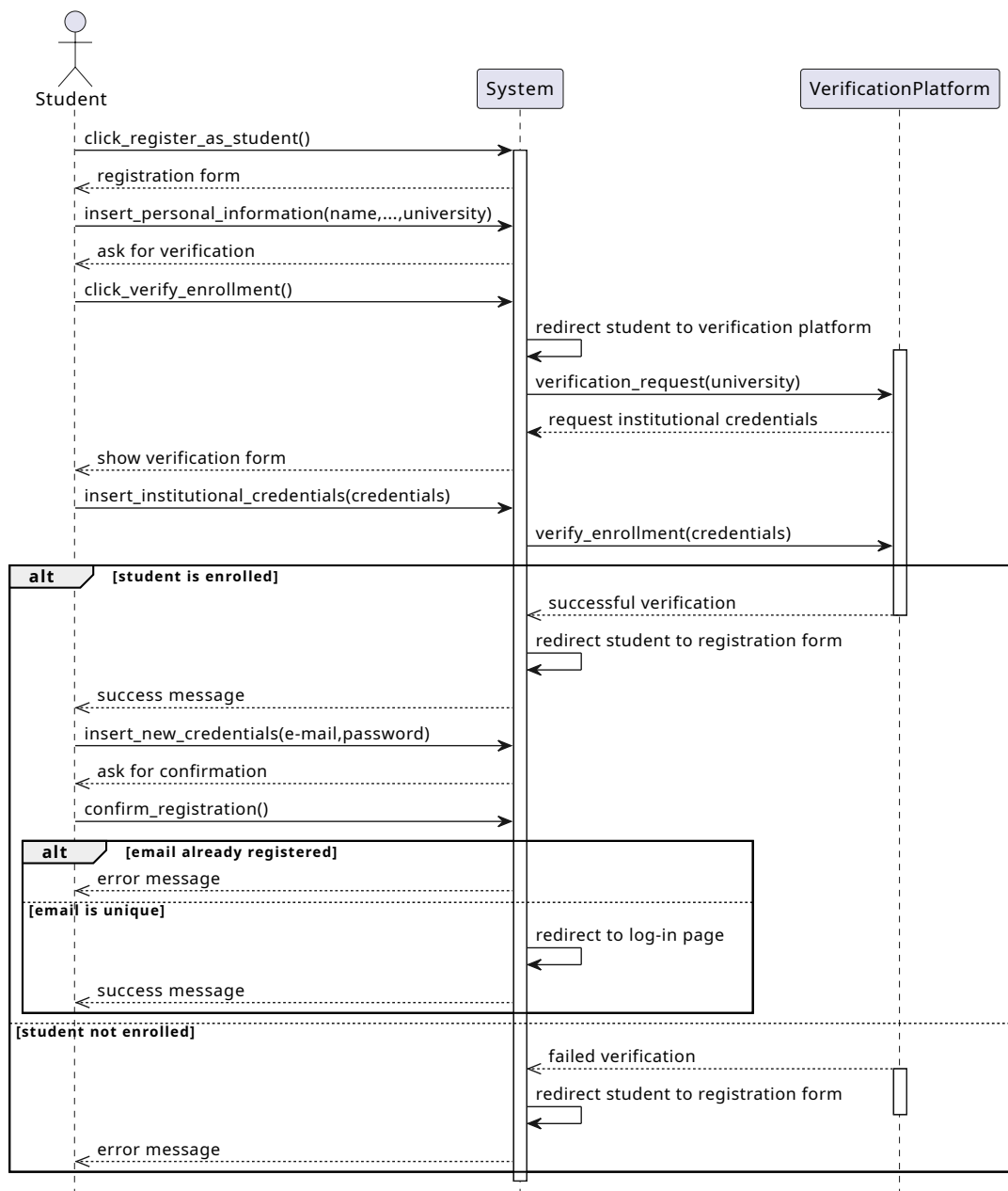
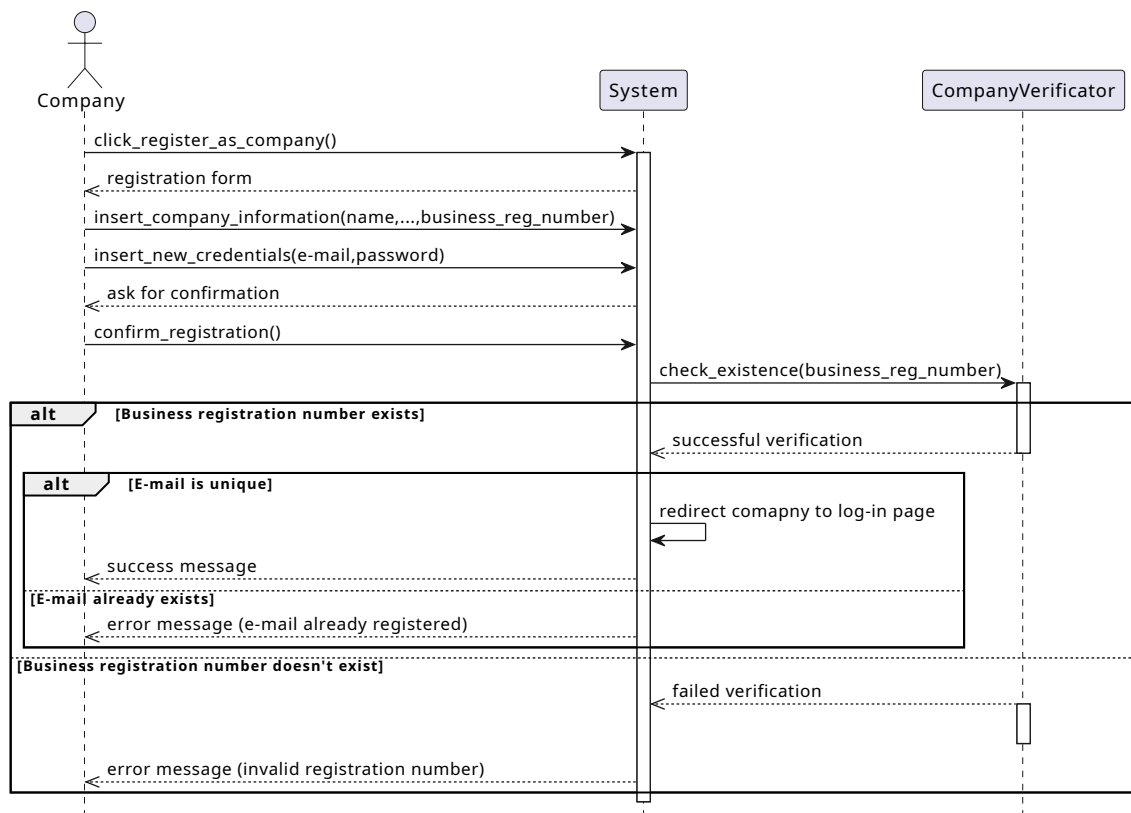


Figure 3.15: Student registration Sequence Diagram

[UC2] - Company registration**Figure 3.16: Company registration Sequence Diagram**

[UC3] - User log-in

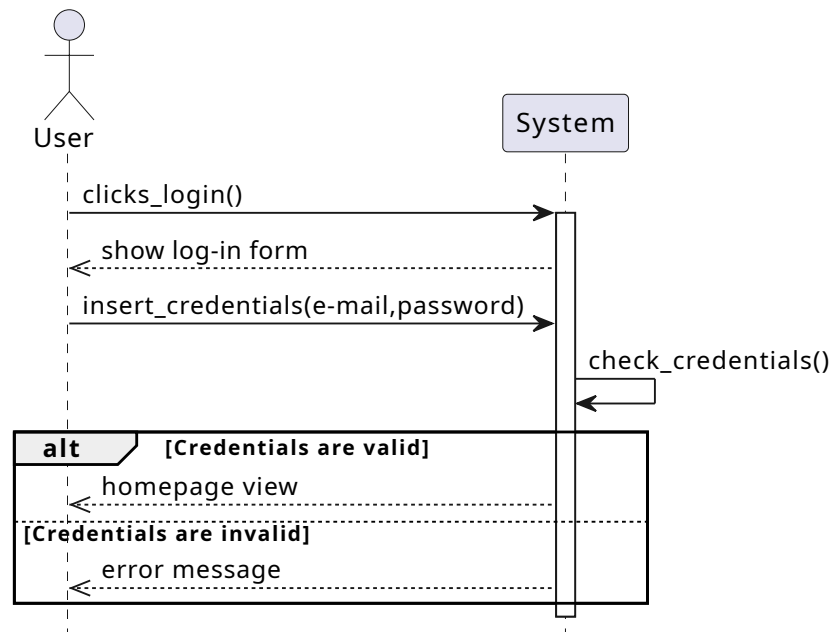


Figure 3.17: User log-in Sequence Diagram

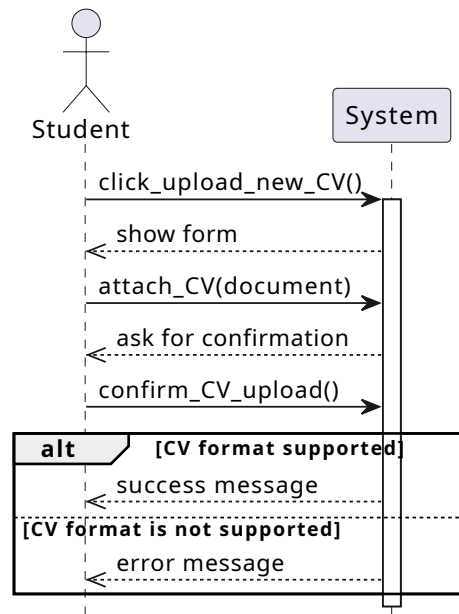
[UC4] - CV upload

Figure 3.18: CV upload Sequence Diagram

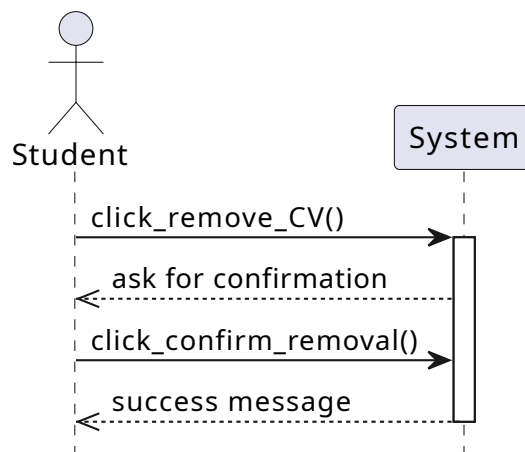
[UC5] - CV removal

Figure 3.19: CV removal Sequence Diagram

[UC6] - Internship publication

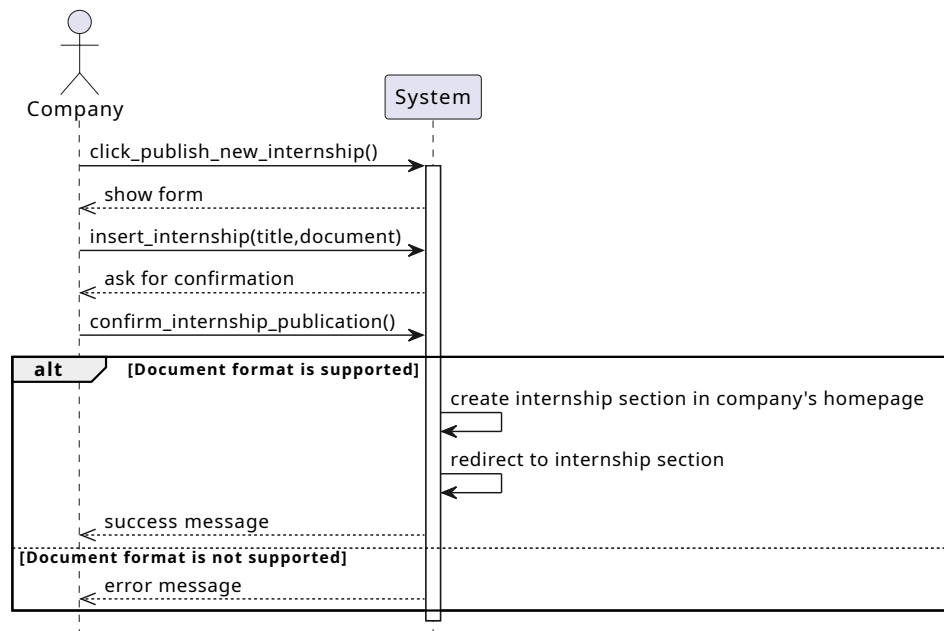


Figure 3.20: Internship publication Sequence Diagram

[UC7] - Internship removal

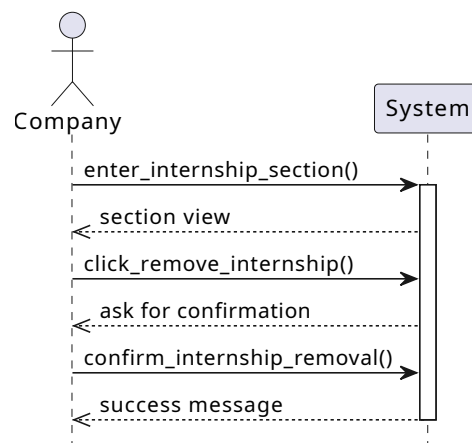


Figure 3.21: Internship removal Sequence Diagram

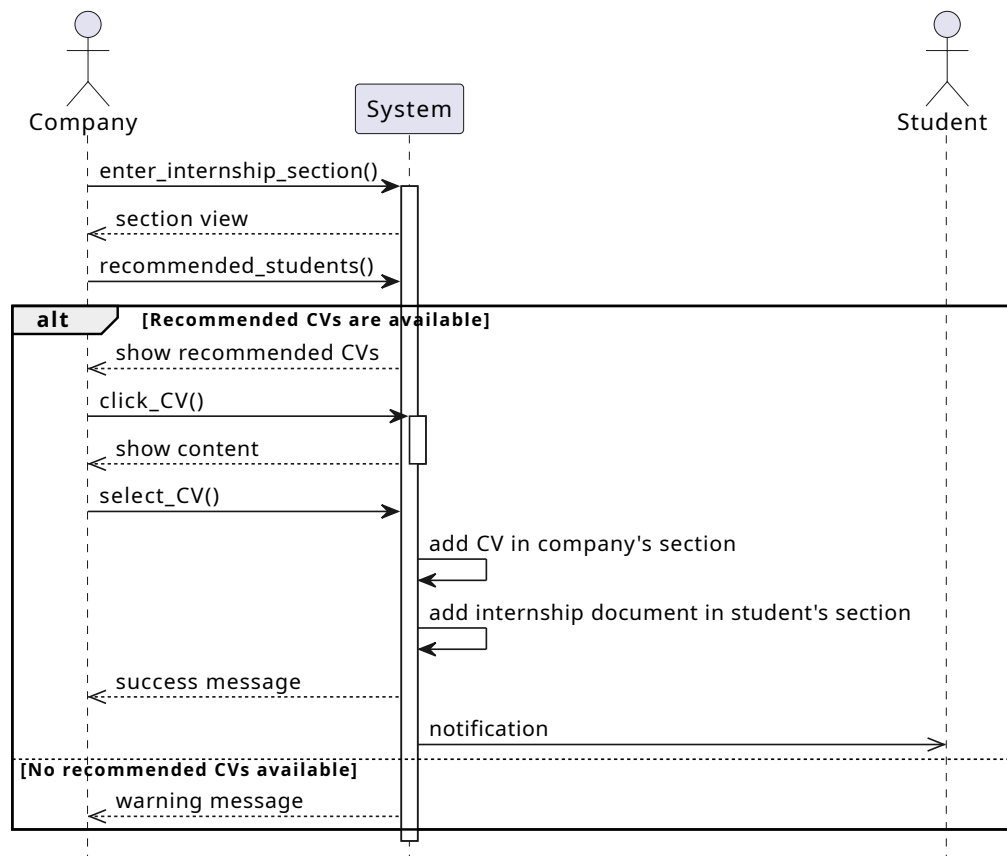
[UC8] - CV selection

Figure 3.22: CV selection Sequence Diagram

[UC9] - Internship selection

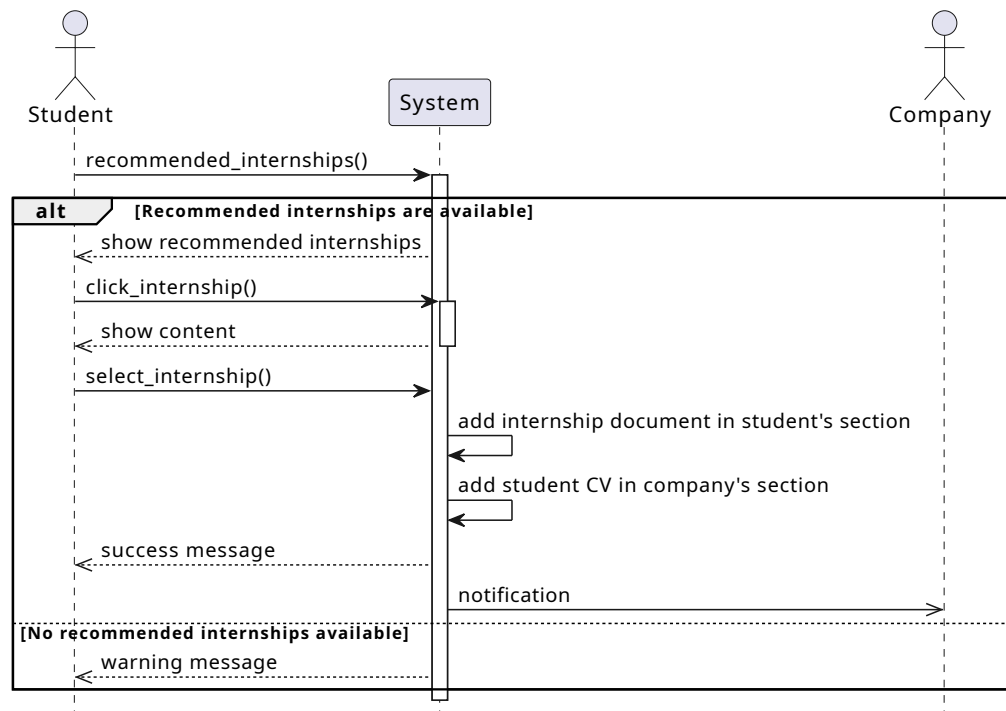
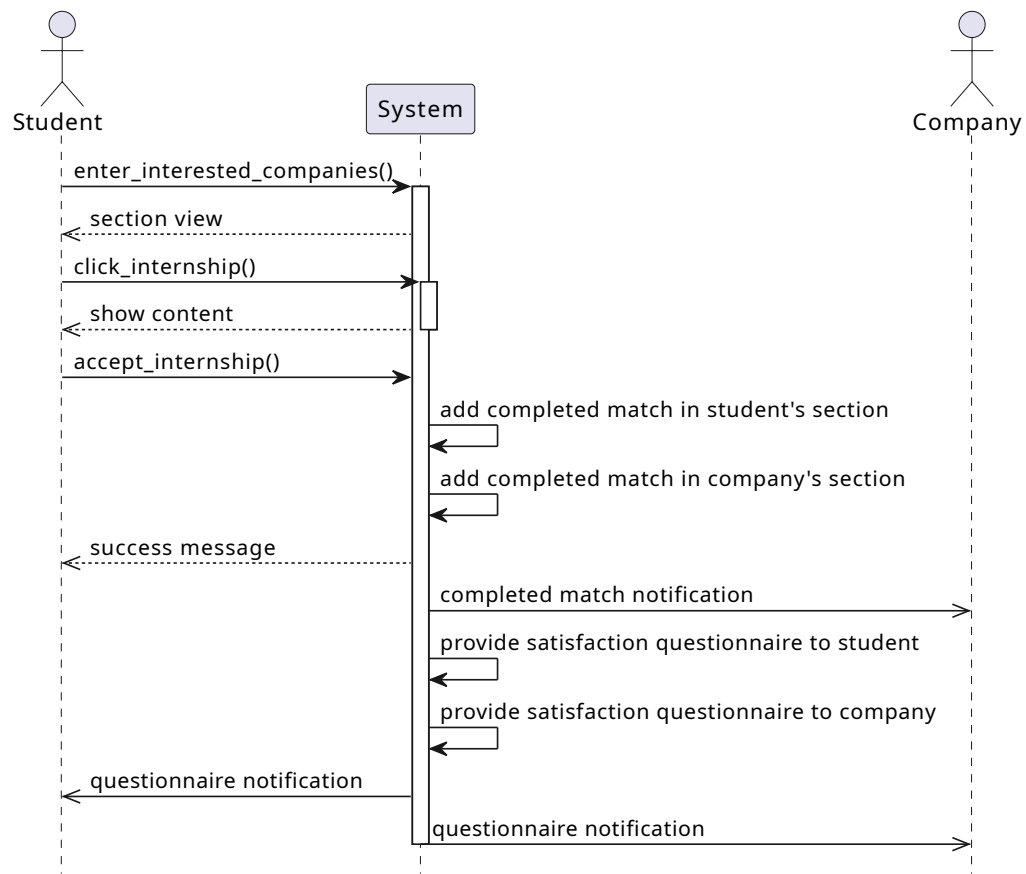


Figure 3.23: Internship selection Sequence Diagram

[UC10] - Match creation by a student**Figure 3.24: Match creation by a student Sequence Diagram**

[UC11] - Match creation by a company

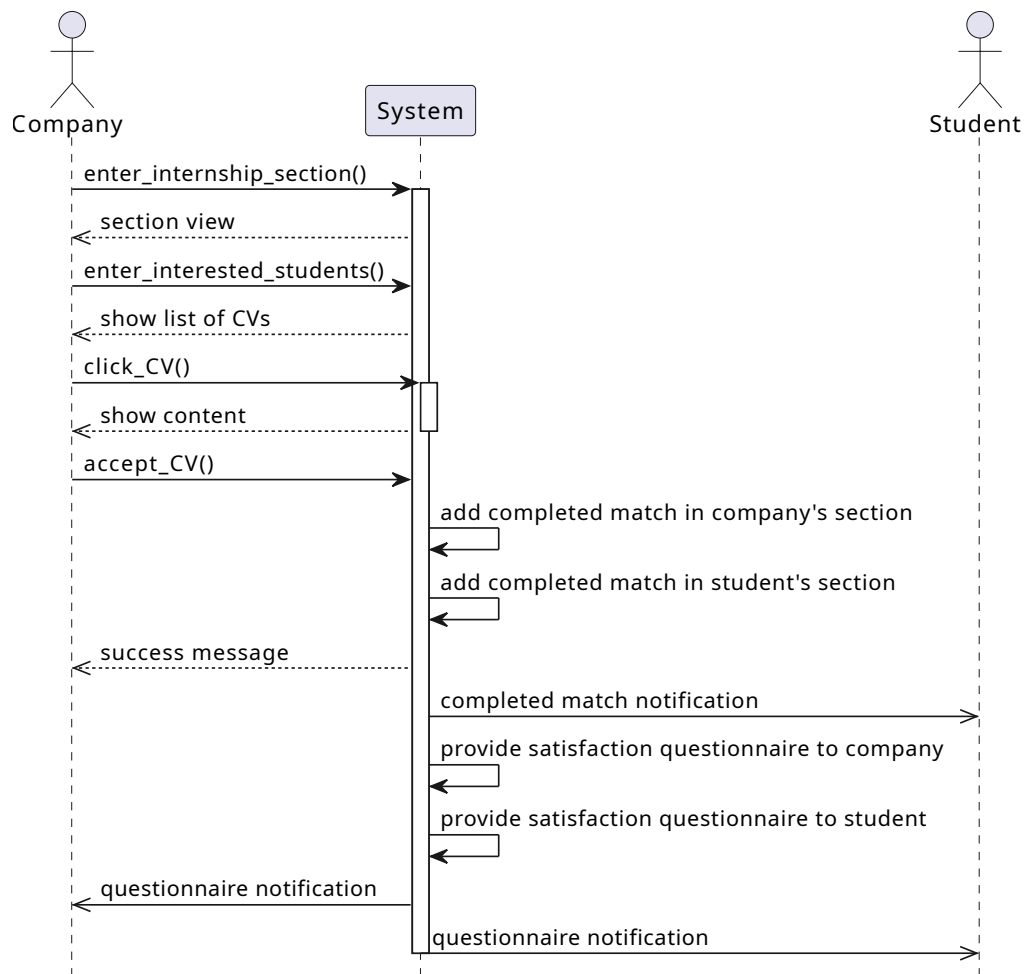


Figure 3.25: Match creation by a company Sequence Diagram

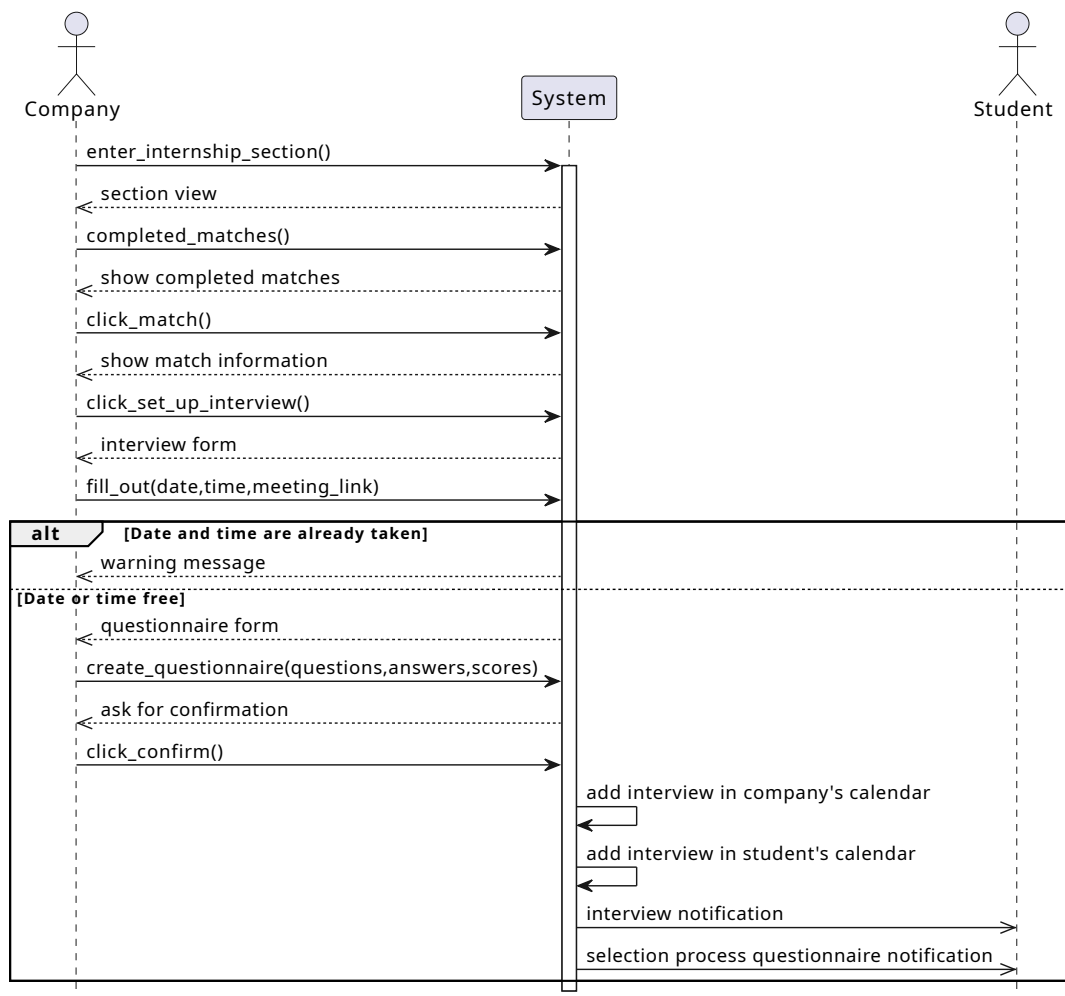
[UC12] - Interview set up

Figure 3.26: Interview set up Sequence Diagram

[UC13] - Selection process questionnaire compilation

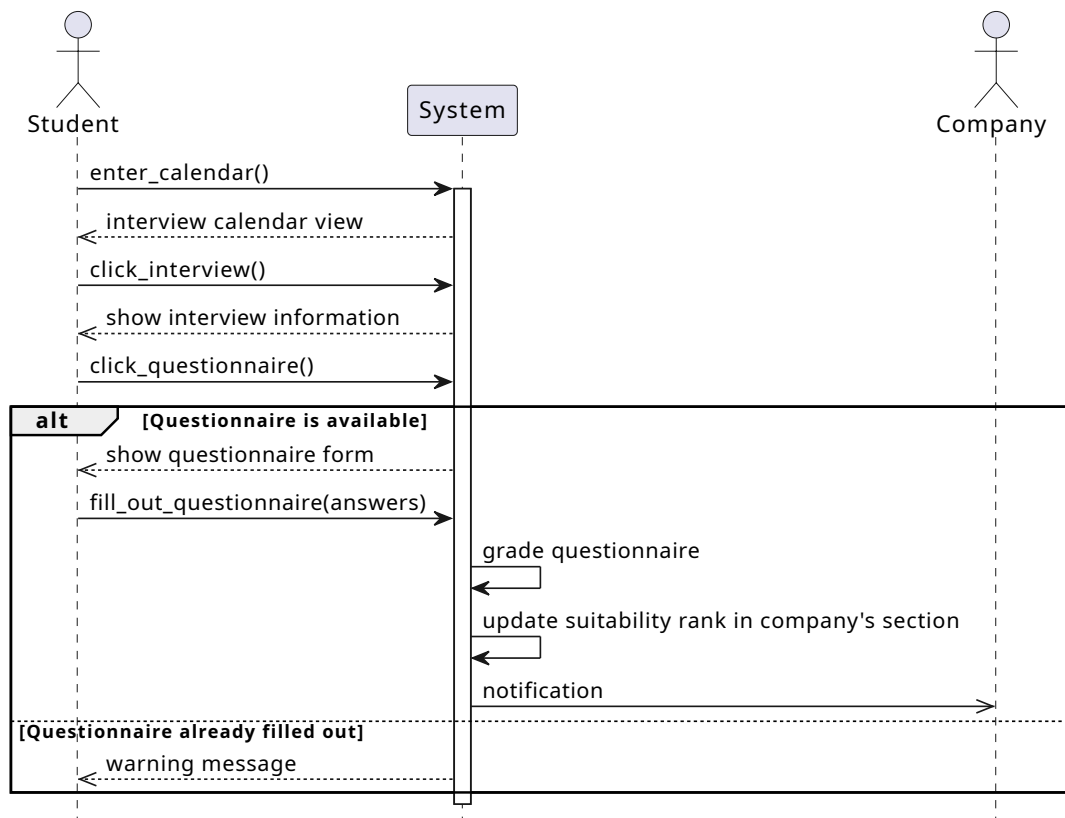
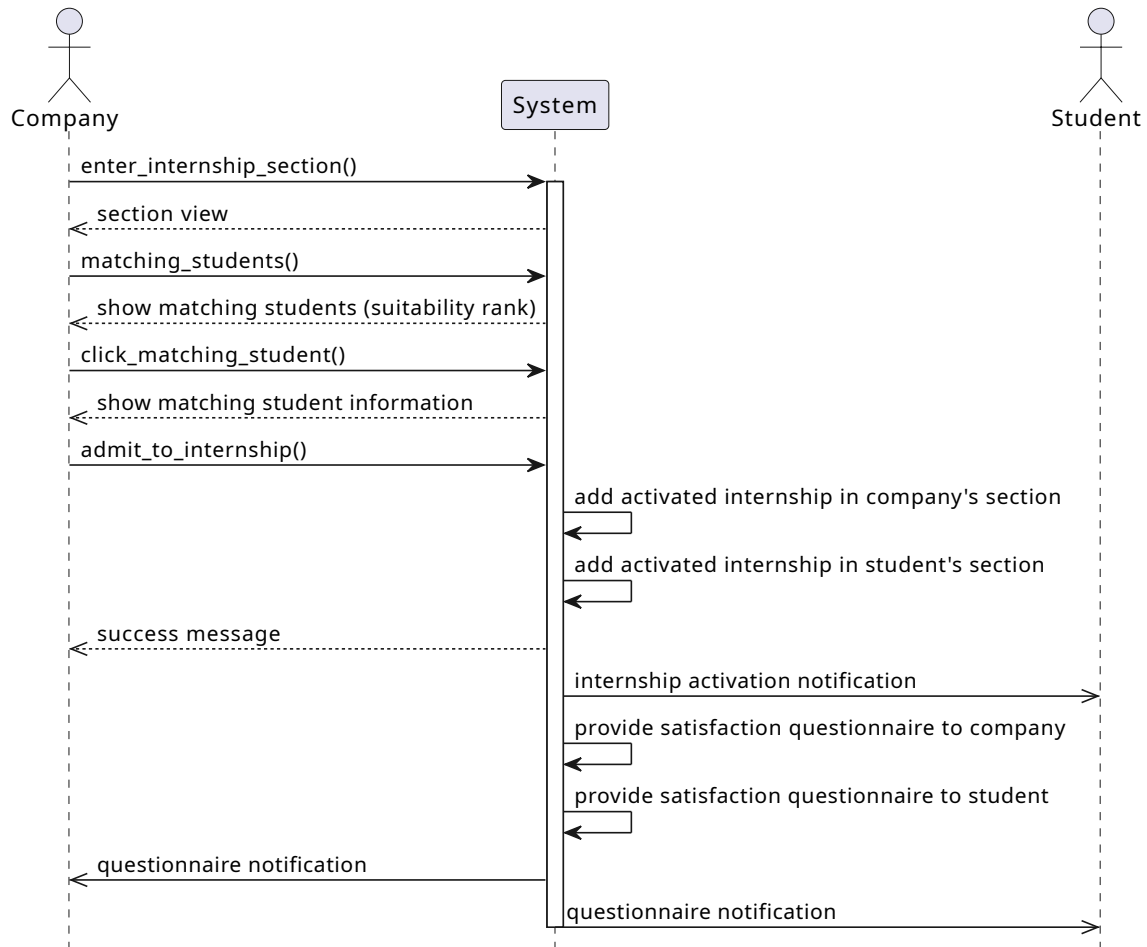


Figure 3.27: Selection process questionnaire compilation Sequence Diagram

[UC14] - Student admission**Figure 3.28: Student admission Sequence Diagram**

[UC15] - Student rejection

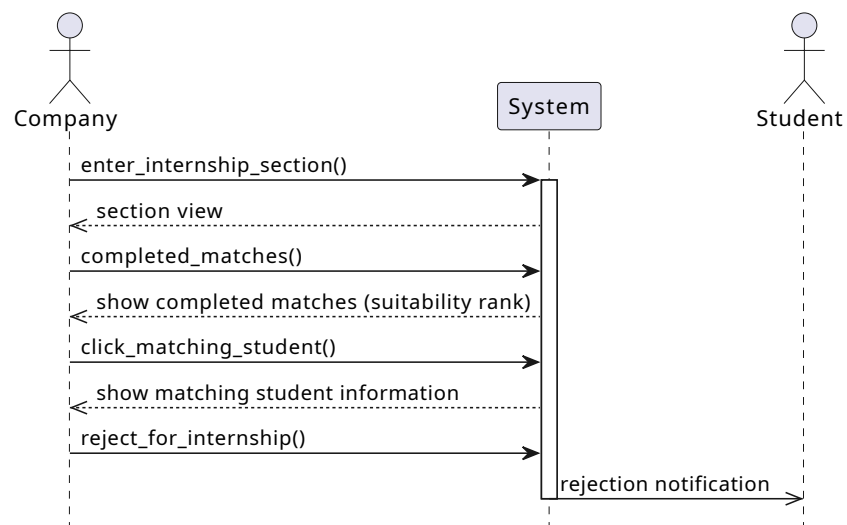


Figure 3.29: Student rejection Sequence Diagram

[UC16] - Student satisfaction questionnaire compilation

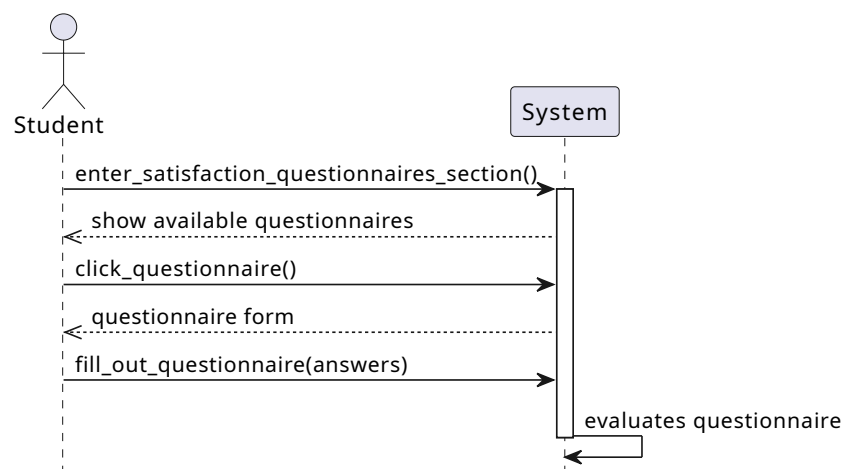


Figure 3.30: Student satisfaction questionnaire compilation Sequence Diagram

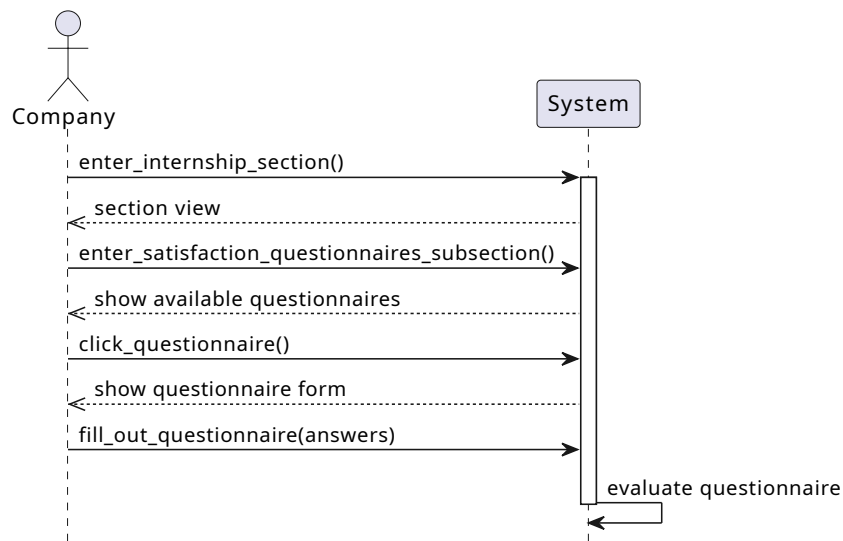
[UC17] - Company satisfaction questionnaire compilation

Figure 3.31: Company satisfaction questionnaire compilation Sequence Diagram

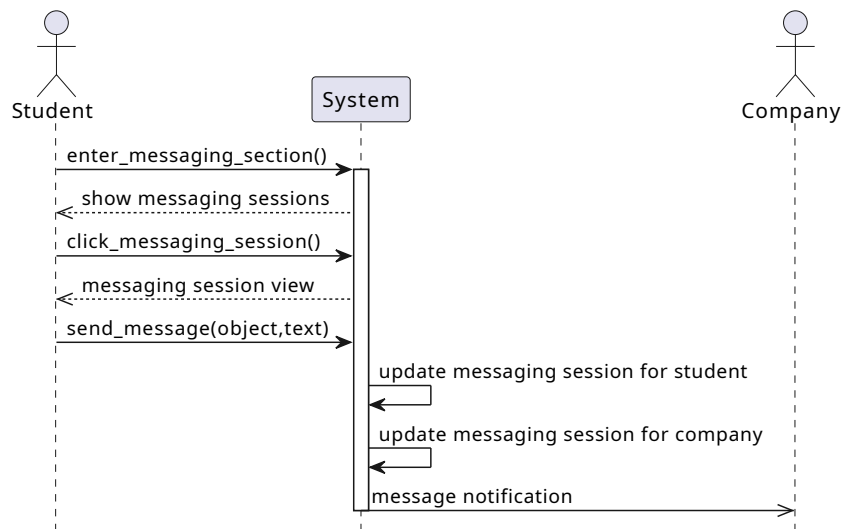
[UC18] - Student message providing

Figure 3.32: Student message providing Sequence Diagram

[UC19] - Company message providing

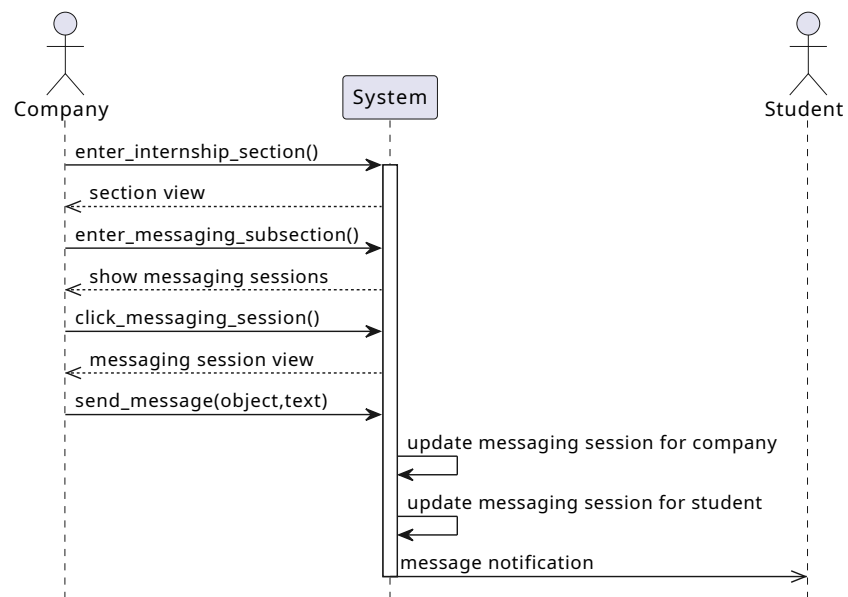


Figure 3.33: Company message providing Sequence Diagram

3.2.4. Requirements Mapping

The following tables (3.20, 3.21, 3.22, 3.23) show the mapping of functional requirements and domain assumptions needed to fulfill the goals.

[G1] Students want to look for available internships that might interest them	
R1, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R22, R23, R24, R25, R42, R43, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54	D1, D2, D3, D4, D5, D7, D8, D9

Table 3.20: Requirement and domain assumption mapping for G1

[G2] Companies would like to look for suitable students' CVs for the internships they offer	
R2, R3, R4, R5, R6, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R42, R43, R45, R46, R47, R48, R49, R50, R51, R52, R53, R55	D1, D2, D3, D4, D5, D7, D8, D9

Table 3.21: Requirement and domain assumption mapping for G2

[G3] Companies would like to admit suitable students to their internships through interviews and structured questionnaires	
R2, R3, R4, R14, R15, R16, R17, R18, R18, R32, R33, R34, R35, R36, R37, R38, R39, R44, R55	D5, D6, D9

Table 3.22: Requirement and domain assumption mapping for G3

[G4] Both students and companies would like to provide information to the other party regarding the matchmaking process and the related internships (such as feedback, suggestions or complaints)	
R1, R2, R3, R4, R20, R21, R40, R41, R54, R55	D5, D9

Table 3.23: Requirement and domain assumption mapping for G4

3.3. Performance Requirements

S&C platform should maintain high performances under both typical and peak usage conditions. The average response time should be lower than 1 second for less critical tasks,

while it should grow until no more than 3 seconds for heavier operations. The system should be able to handle at least 1000 concurrent users on average, and, at the same time, should include mechanisms to increase this number up to 10000 during high-request periods. Also, the expected number of registered users in the initial phase is around 1 million, even if the system should be able to support a doubling of this measure.

3.4. Design Constraints

3.4.1. Standards Compliance

As mentioned before, the system must follow GDPR for user personal data treatment. Additionally, it should align with most recent versions of web standards, such as HTML5, CSS3 and W3C Responsive Web Design, to adapt the application to different screen sizes and devices. As concerns accessibility, the platform should comply with the Web Content Accessibility Guidelines (WCAG) level AAA, to ensure access and usability to all kinds of users.

3.4.2. Hardware Limitations

Given the web-based nature of S&C, the only hardware constraint for users will be having a device (desktop or mobile) equipped with a web browser and a functioning Internet connection.

3.5. Software System Attributes

Additional non-functional requirements of the platform are listed below.

3.5.1. Reliability

The system should have a Mean Time Between Failures (MTBF) of 1000 hours, ensuring high level reliability. Fault detection and recovery mechanisms should therefore be implemented.

3.5.2. Availability

S&C system is expected to reach an availability of at least 99.9% with a down time of 8.76 hours/year, excluding maintenance, which should be planned in advance with users being notified.

3.5.3. Security

To ensure security, the system should utilize HTTPS protocol, with TLS data encryption. Moreover, to preserve user information confidentiality and integrity, passwords should be hashed and a strict access control to data must be adopted. Finally, also external APIs security has to be assessed.

3.5.4. Maintainability

In order to support easy modification, either updates or fixes, the system should meet the following maintainability requirements. The code should be well documented and easily readable and understandable; modularity, together with low coupling and high cohesion, should be prioritized; version control and automated testing should be adopted throughout the whole process.

3.5.5. Portability

Being a web-based application, S&C should achieve a high level of portability. It should therefore be compatible and optimized for the majority of modern web browsers (such as Google Chrome, Microsoft Edge, Mozilla Firefox and Safari) and for all desktop and mobile devices. In addition to that, the system should be platform independent, supporting most popular operating systems (Linux, Windows, macOS).

4 | Formal Analysis Using Alloy

In this Chapter, the Alloy specification of the main entities and constraints comprising the application domain is presented. The aim of the model is to specify the domain consistently with the Class Diagram in Section 2.1. For the sake of simplicity, the process of CV selection and upload and the process of internship selection and publication have not been considered in the specification; therefore, the related relationships have been specified as immutable. Instead, within the scope of dynamic analysis, particular attention has been devoted to the processes of fill-out and evaluation of satisfaction questionnaires and selection process questionnaires.

4.1. Signatures

The following Alloy model shows the main entities and relations that hold in the application domain.

```

1 open util / relation
2 open util / boolean
3
4 // User
5 abstract sig User {}
6
7 // Student
8 sig Student extends User {
9     cv: one CV,                      // associated CV
10    selectedInternships: set Internship, // selected internships
11    interestedCompanies: set Internship,
12    matches: set Match,               // completed matches
13 }
14
15 // Company
16 sig Company extends User {
17     internships: set Internship,      // published internships

```

```

18   selectedCVs: Internship -> set CV,      // selected CVs for each
      internship
19   interestedStudents: Internship -> set CV,
20   matches: set Match                      // completed matches
21 }
22
23 // Student CV
24 sig CV {
25   owner: one Student                      // CV's owner
26 }
27
28 // Internship
29 sig Internship {
30   publisher: one Company                  // Internship's publisher
31 }
32
33 // Questionnaire
34 abstract sig Questionnaire{
35   var compiled: one Bool
36 }
37
38 // Questionnaire for the selection process
39 sig SelectionQuestionnaire extends Questionnaire{
40   var grade: one Int,
41   interview: one Interview                // related interview
42 } {grade >= 0 and grade <= 10}
43
44 // Satisfaction questionnaire
45 abstract sig SatisfactionQuestionnaire extends Questionnaire{
46   var evaluation: one Bool
47 }
48
49 // Satisfaction questionnaire for matchmaking process
50 sig MatchingQuestionnaire extends SatisfactionQuestionnaire{
51   match: one Match                       // related match
52 }
53
54 // Satisfaction questionnaire for activated internship
55 sig InternshipQuestionnaire extends SatisfactionQuestionnaire{

```

```

56     internship: one ActiveInternship           // related active
           internship
57 }
58
59 // Completed match
60 sig Match {
61     student: one CV,           // matching CV
62     internship: one Internship, // matching internship
63     questionnaires: some MatchingQuestionnaire, // associated
           questionnaires
64     activeInternship: lone ActiveInternship // activated internship
65 }
66
67 // Interview
68 sig Interview {
69     match: one Match,           // related match
70     questionnaire: one SelectionQuestionnaire // questionnaire
           for selection process
71 }
72
73 // Active internship
74 sig ActiveInternship {
75     match: one Match,           // related match
76     questionnaires: some InternshipQuestionnaire, // associated
           questionnaires
77     messages: set Message      // messaging session
78 }
79
80 // Message
81 sig Message {
82     sender: one Student + Company, // company/student who
           sent the message
83     receiver: one Student + Company, // company/student who
           received the message
84     activeInternship: one ActiveInternship // active
           internship associated to the messaging session
85 }

```

4.2. Constraints

This section presents the Alloy specification of the main constraints that must be satisfied within the model described above. They ensure that the results produced by the Alloy analyzer are consistent with the specifications outlined so far in the RASD.

```

1  // Ensures that the student associated to a CV has actually
    uploaded that CV, and
2  // that the company associated to an internship has actually
    published that internship
3  fact Ownership {
4      (all c: CV, s: Student |
5          c = s.cv iff s = c.owner)
6      and
7      (all i: Internship, c: Company |
8          i in c.internships iff c = i.publisher)
9  }
10
11 // Ensures that if an internship is in the "interestedCompanies"
    list of a student, their
12 // CV must be in the "selectedCVs" list of the corresponding
    company for that internship
13 fact InterestedCompanyImpliesSelectedCV {
14     all s: Student, i: Internship |
15         i in s.interestedCompanies iff s.cv in
            i.publisher.selectedCVs[i]
16 }
17
18 // Ensures that if a student CV is in the "interestedStudents"
    list of a company for a
19 // specific internship, the internship must be in the
    "selectedInternships" list
20 // of the corresponding student
21 fact InterestedStudentImpliesSelectedInternship {
22     all co: Company, c: CV, i: Internship |
23         c in co.interestedStudents[i] iff i in
            c.owner.selectedInternships
24 }
25
26 // Ensures that the student and company involved in a match have

```



```

    actually that match
27 // in their completed matches list
28 fact CorrespMatchUsers {
29     (all m: Match, s: Student |
30         m in s.matches iff s = m.student.owner)
31     and
32     (all m: Match, c: Company |
33         m in c.matches iff c = m.internship.publisher)
34 }
35
36 // Ensures that there cannot be two matches associated to the
    same pair
37 // CV-internship
38 fact NoDuplicatedMatches {
39     all m1, m2: Match |
40         (m1.student = m2.student and m1.internship =
41             m2.internship)
42         implies m1 = m2
43 }
44
45 // Ensures that every match is associated to exactly 2
    satisfaction questionnaires
46 // (one for the student and one for the company) and that these
    questionnaires are
47 // actually related to that match
48 fact correspMatchQuest {
49     all m: Match, q: MatchingQuestionnaire |
50         (q in m.questionnaires iff q.match = m)
51         and
52         (#m.questionnaires = 2)
53 }
54
55 // Ensures that every active internship is associated to exactly
    2 satisfaction questionnaires
56 // (one for the student and one for the company) and that these
    questionnaires are
57 // actually related to that active internship
58 fact correspInternshipQuest {

```

```

59     all a: ActiveInternship, q: InternshipQuestionnaire |
60         (q in a.questionnaires iff q.internship = a)
61         and
62         (#a.questionnaires = 2)
63 }
64
65 // Ensures that an active internships deriving from a match is
66 // actually associated
67 // to that match
68 fact correspMatchActiveInt {
69     all a: ActiveInternship, m: Match |
70         a = m.activeInternship iff m = a.match
71 }
72
73 // Ensures that a selecetion process questionnaire is associated
74 // to the correct
75 // interview
76 fact correspInterviewQuest {
77     all q: SelectionQuestionnaire, i: Interview |
78         q = i.questionnaire iff i = q.interview
79 }
80
81 // Ensurerere that for each match only one interview can be set up
82 fact NoMultipleInterviews {
83     all i1, i2: Interview |
84         (i1.match = i2.match) implies (i1 = i2)
85 }
86
87 // Ensures that a message related to an active internship is
88 // contaied in the
89 // messaging session of that internship
90 fact correspMexActiveInt {
91     all m: Message, a: ActiveInternship |
92         m in a.messages iff m.activeInternship = a
93 }
94
95 // Ensures that a message is sent by a student and received by a
96 // company (or viceversa),
97 // and that the 2 parties are actually involved in the related

```

```

    active internship
94 fact MexConsistency {
95     all m: Message | ((m.sender =
        m.activeInternship.match.student.owner) and
96         (m.receiver =
            m.activeInternship.match.internship.publisher))
97     or
98     ((m.sender =
        m.activeInternship.match.internship.publisher)
        and
99     (m.receiver =
        m.activeInternship.match.student.owner))
100 }
101
102 // Ensures that if a satisfaction questionnaire has not been
    compiled, then it doesn't contribute to the
103 // recommendation process.
104 // Instead, once the questionnaire has been compiled, the
    evaluation must never change.
105 fact EvaluationConsistency {
106     all q: SatisfactionQuestionnaire |
107         always(q.compiled = False implies q.evaluation =
            False)
108         and
109         always (q.compiled = True implies always
            q.evaluation = q.evaluation')
110 }
111
112 ' // Ensures that if a selection process questionnaire has not
    been compiled, the related match will
113 // be at the bottom of the suitability rank of the corresponding
    company, because no grade has been
114 // established yet.
115 // Instead, once the questionnaire has been compiled, the grade
    must
116 // never change.
117 fact GradeConsistency {
118     all q: SelectionQuestionnaire |
119         always (q.compiled = False implies q.grade = 0)

```

```

120         and
121         always (q.compiled = True implies always q.grade =
              q.grade')
122     }
123
124     '// Ensures that once a questionnaire has been compiled, it can
        never be compilable again
125 fact CompilationConsistency {
126     all q: Questionnaire |
127         always(q.compiled = True implies after always q.compiled
              = True)
128 }

```

4.3. Examples

In Figure 4.1 we can see a simple example of world generated by the alloy analyzer through the run operation below. In particular, the example highlights the concept of match creation, which correctly involves one student CV and one internship and is related to two satisfaction questionnaires.

```

1 pred show {}
2 run show for 3 but exactly 1 Match, exactly 1 Internship

```

In Figures 4.2 and 4.3 more complex examples are shown. In particular, the first one represents the activation of an internship starting from a completed match between a CV and an internship (two satisfaction questionnaires are correctly associated to the active internship). The second one shows also an example of communication between a company and a student in the context of an active internship; the communication consists in two messages, both sent by the student and received by the company. These two worlds are generated by the analyzer by running the following operations.

```

1 pred show {}
2 run show for 7 but exactly 1 Match, exactly 1 Internship,
    exactly 1 Student, exactly 1 ActiveInternship, exactly 0
    Message

```

```

1 pred show {}
2 run show for 7 but exactly 1 Match, 1 Internship, 1 Student, 1
    ActiveInternship

```

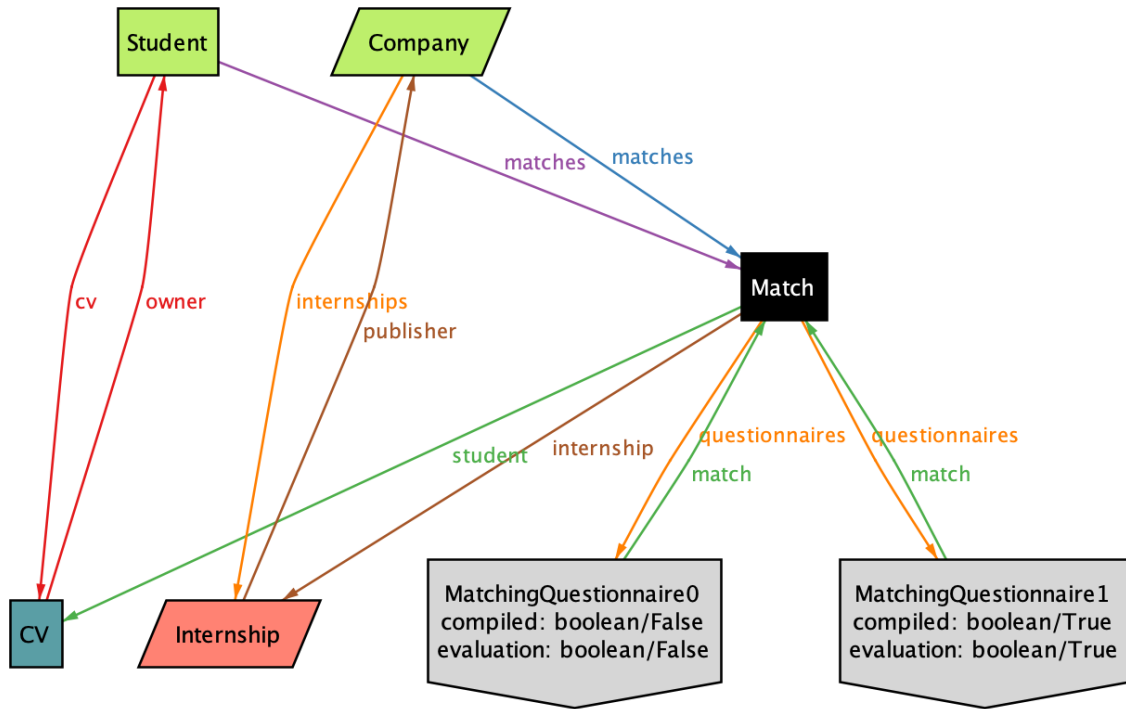


Figure 4.1: World example: completed match

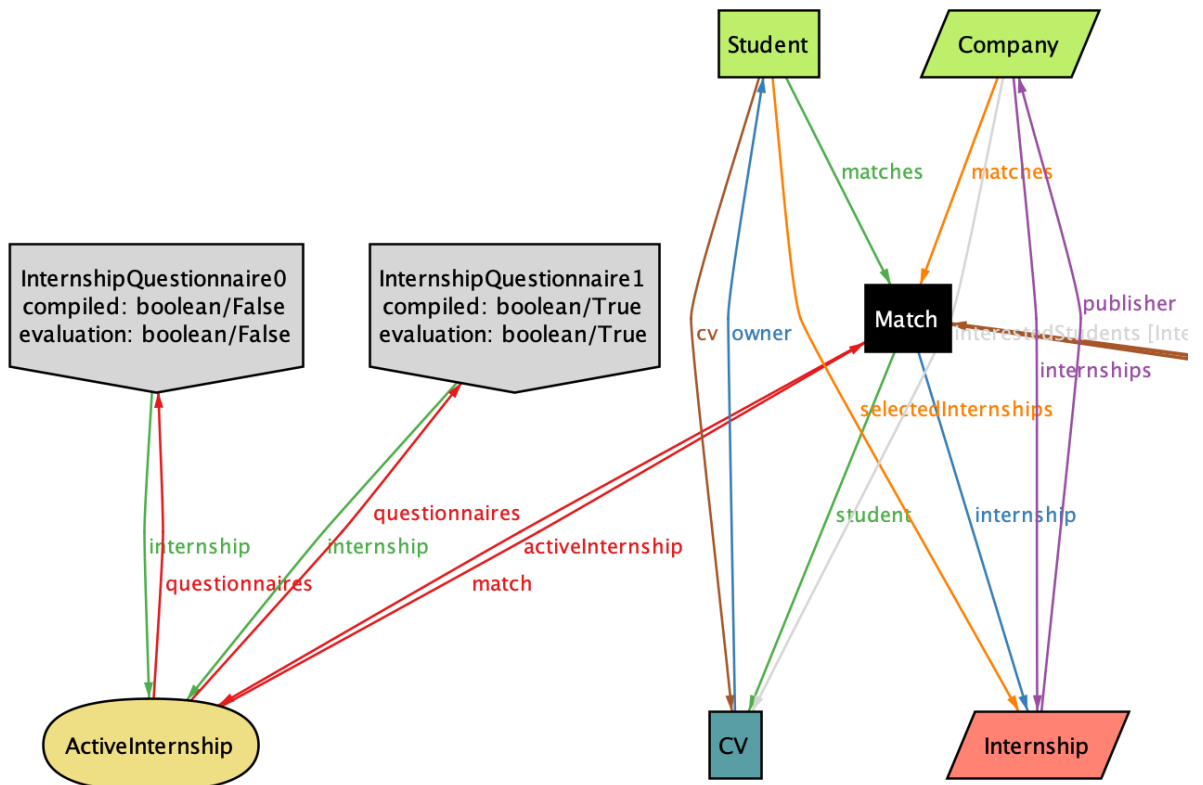


Figure 4.2: World example: active internship

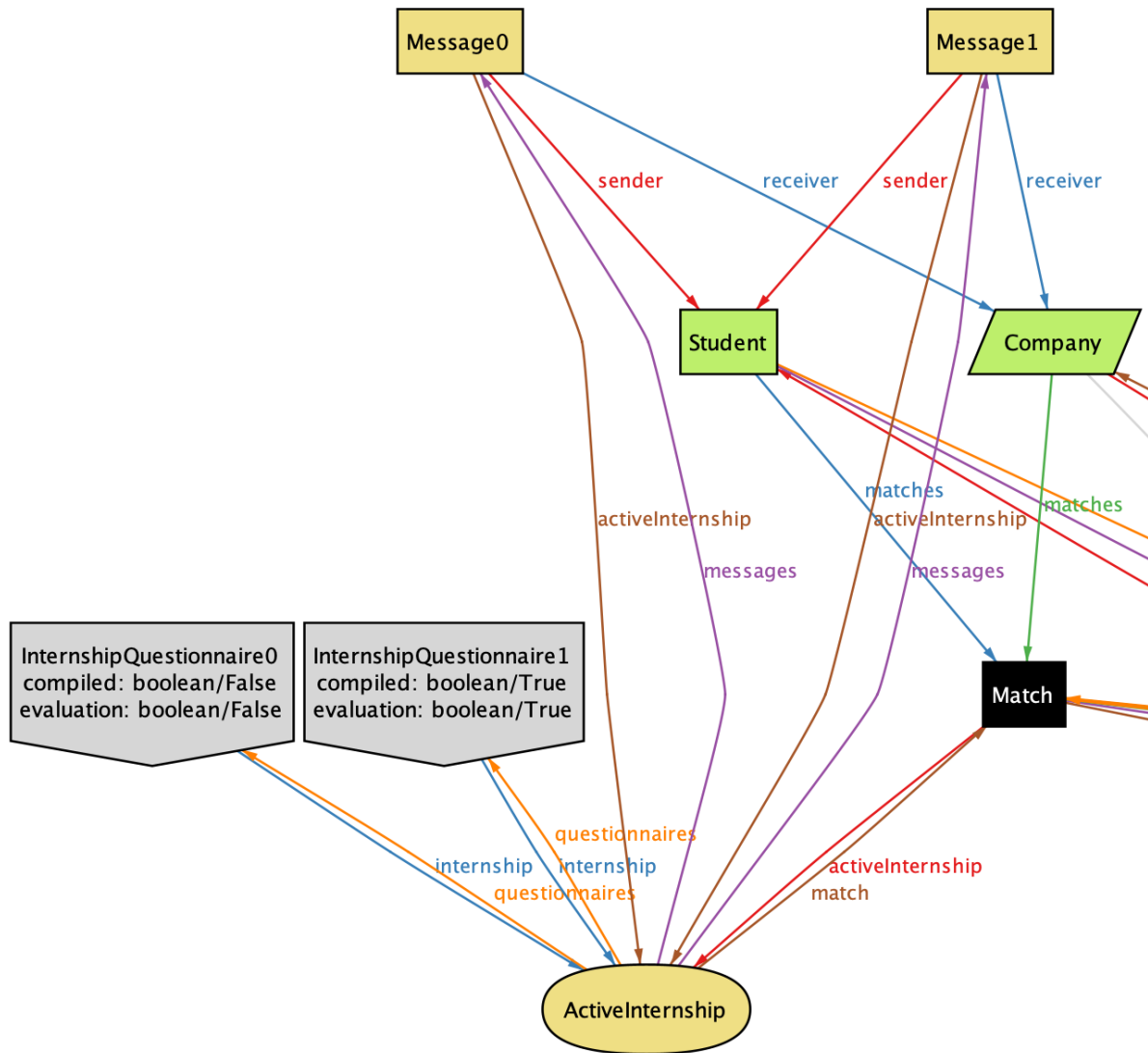


Figure 4.3: World example: message communication

In the following Figures an example of dynamic analysis of the Alloy model is described. It shows the compilation of a selection process questionnaire in the context of an interview. In the first instant (Figure 4.4), the questionnaire still hasn't been filled out and its grade is correctly set to 0, while in the second one (Figure 4.5) the questionnaire is compiled and its grade changes.

```

1 pred SelQuestCompilation[q: SelectionQuestionnaire]{
2     q.compiled = False; q.compiled = True
3 }
4 run SelQuestCompilation for 10 Int

```

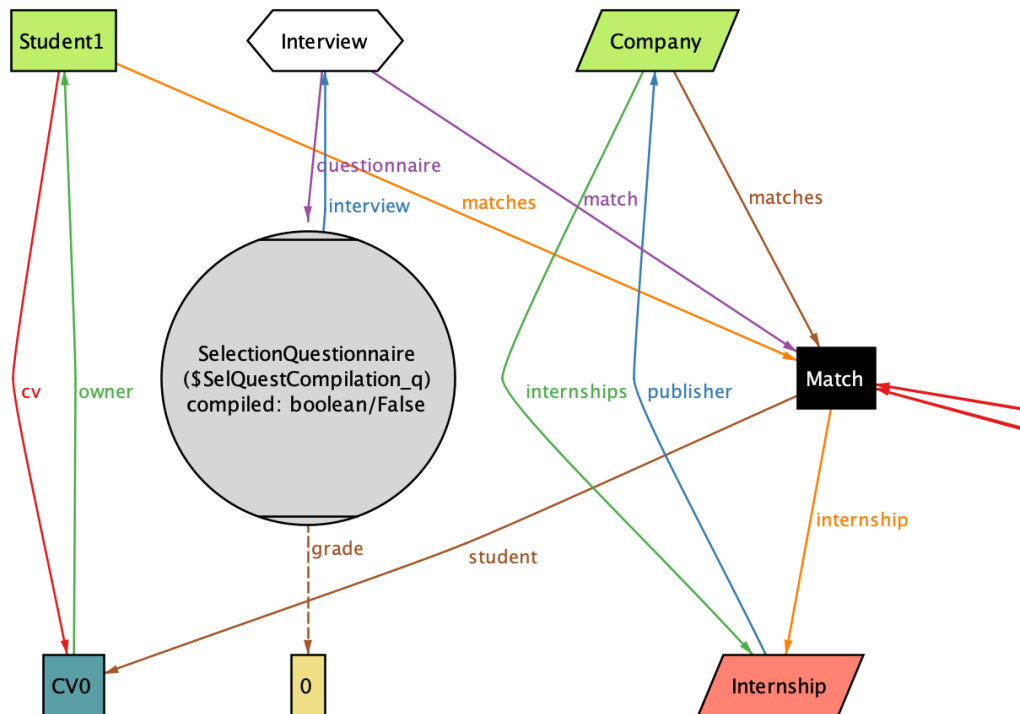


Figure 4.4: World example: selection process questionnaire (step 1)

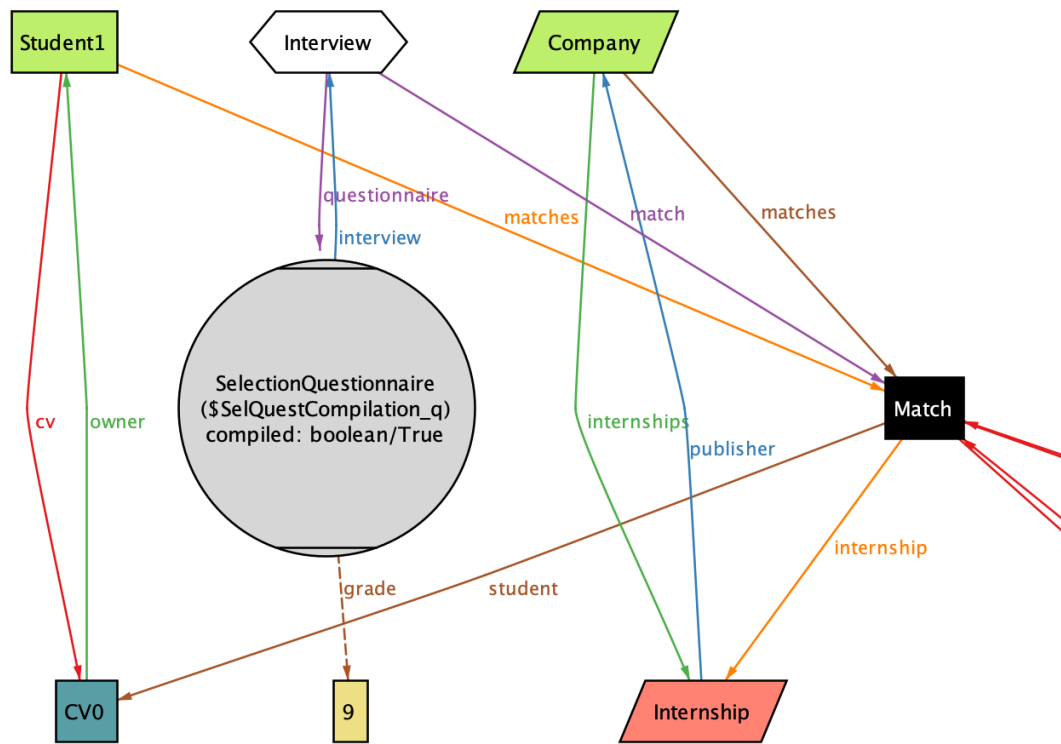


Figure 4.5: World example: selection process questionnaire (step 2)

To conclude the analysis of the model, we considered two assertions regarding the questionnaire evaluation, in order to find counterexamples that could highlight interesting properties of the application.

In the specification of the constraints, we ensured that a not compiled satisfaction questionnaire does not contribute to the recommendation process, therefore its evaluation must be set to False. With the following assertion it's shown that a satisfaction questionnaire can have a negative evaluation not only when the questionnaire is yet to be filled out, but also when the system has negatively evaluated the user's answers.

```

1 assert NoNegativeEvaluation { always
2     (no q: SatisfactionQuestionnaire |
3         q.compiled = False and after (q.compiled = True and
4             q.evaluation = False))
5 }
6 check NoNegativeEvaluation

```

Executing "Check NoNegativeEvaluation"

Solver=sat4j Steps=1..10 Bitwidth=4 MaxSeq=4 SkolemDepth=1 Symmetry=20 Mode=batch
 1..2 steps. 11267 vars. 780 primary vars. 18674 clauses. 126ms.
Counterexample found. Assertion is invalid. 22ms.

Figure 4.6: Counterexample for assertion (1)

The second assertion shows an interesting and not trivial aspect of the user-system interaction. The selection process questionnaire evaluation is a mechanism that helps companies admit students for an internship, thanks to the creation of a suitability rank of matching students for that internship. However, it mustn't be mandatory for companies to admit only the higher ranked students. They should be free to admit or reject them based also on other aspects of the selection process (such as the actual interview carried out through a meeting platform). The following assertion check shows that a company can admit a student whose match is not at the top of the suitability rank for a specific internship.

```

1 assert AdmitOnlyTopStudent { always
2     (no q1, q2: SelectionQuestionnaire |
3         q1 != q2
4         and
5         q1.grade > q2.grade
6         and

```

```
7         q1.interview.match.activeInternship = none)
8     }
9     check AdmitOnlyTopStudent for 10 but exactly 1 Company, exactly 1
10        Internship, exactly 2 Student, exactly 2 Match,
    exactly 1 ActiveInternship, exactly 2 Interview, 10 Int
```

Solver=sat4j Steps=1..10 Bitwidth=10 MaxSeq=10 SkolemDepth=1 Symmetry=20 Mode=batch
1..1 steps. 385683 vars. 10634 primary vars. 1340227 clauses. 1590ms.
Counterexample found. Assertion is invalid. 5336ms.

Figure 4.7: Counterexample for assertion (2)

5 | Effort Spent

This is the effort spent chapter.

6 | References

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